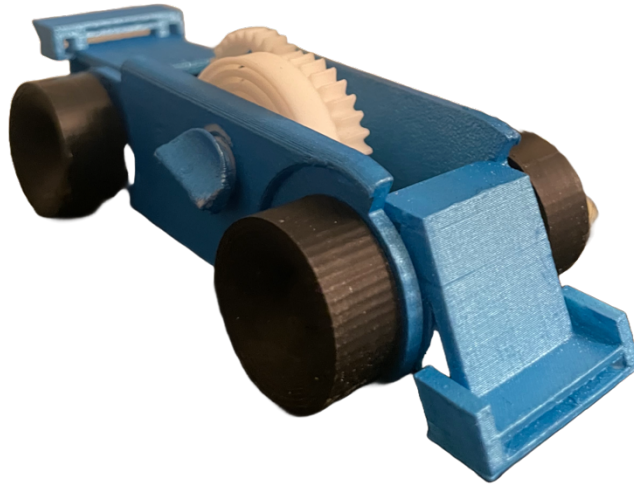




Blueprint Design Report

Report Created W25, with contributions from all team members

Course (ENGG*2100)

Group 63

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Executive Summary

This report outlines the complete design process and evaluation of Blueprint 1, a wind-up Formula 1 style toy car developed to meet the requirements of the ENGG*2100 Winter 2025 design project. The goal of the project was to create a safe, fun, and functional toy that could be fully enclosed within a Kinder™ Surprise Egg. The toy was required to be safe, visually engaging, mechanically sound, and easy to assemble without the use of tools or adhesives.

The design process was guided by clearly defined user needs, including reliable motion, durable materials, and visual appeal, as well as strict constraints on size, manufacturability, and safety. A spring driven wind-up mechanism was selected as the propulsion system. Various spring sizes were evaluated through prototyping to determine the optimal balance between energy storage and compactness. Careful consideration was given to gear placement, body sizing and wheel design to ensure consistent reliable performance.

The final product was manufactured using PLA filament for all printed components, with a total print weight of 70.06 grams. Rubber bands were fitted to the wheels to improve traction on smooth surfaces and was finished with a blue metallic paint. The toy was designed to be assembled without tools in a series of snap fit steps. No adhesives or fasteners were required, simplifying both production and user interaction.

From a project management standpoint, the development process remained within the prototype budget. The final version of the toy cost approximately \$2.30, including filament, rubber bands, and paint. An earlier prototype, which exceeded size limitations and underperformed in distance tests, led to a redesign of the spring. Early challenges with team scheduling were resolved using a shared calendar system, which improved coordination and allowed the group to meet project deadlines effectively.

In all, Blueprint 1 successfully meets the functional, dimensional and performance-based objectives outlined at the start of the project. This report details the design methodology, evaluation process, manufacturing approach, assembly instructions (Appendix C), and overall project management strategies that led to the final outcome.

Statement of Contributions

*****EACH GROUP MEMBER MUST SIGN BELOW TO RECEIVE THEIR MARKS*****

By signing this form, using a signature (not typed – suggest using “fill and sign” in adobe after generating the final .pdf version of the report), each member is stating that they made a significant contribution to the preparation of the contents within (e.g. research, design work), and to the writing of this report, and that the distribution of effort listed below is accurate.

Required statement: Each of the students listed below has read and agrees with the contents of this report. Additionally, each student has made significant contributions (listed below) to the preparation of this report. *If you do not feel comfortable with this statement, please contact your instructor and explain the situation.*

Name (Printed)	Signature	Design Contributions & Sections Written or Edited
Avneet Aujla		- Executive Summary - Introduction - Design Evaluation - Assembly Instructions Project Management
Adib Mazloumi		- Constraints & Criteria - Design Process - Design Defense
Saqsham Chopra		- Marketing Strategy Manufacturing & Cost Analysis 2D Drawings
Faran Saharan		- Proposed Design Conclusion & Recommendations 2D Drawings Formatting document

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1 Introduction

1.1 Problem Definition

In today's toy market, there is a growing demand for engaging and interactive products that not only entertain but also encourage learning. Traditional battery powered toy cars, while popular, come with drawbacks such as limited user engagement, environmental concerns, and ongoing costs for battery replacements. To address these issues our team at Blueprint designed a Formula 1 inspired wind-up toy race car that operates purely on mechanical energy, eliminating the need for batteries. This toy is intended for children aged seven and older who enjoy hands on play and for parents seeking educational toys that introduce fundamental engineering concepts in an accessible way. The design prioritizes durability, functionality and realism ensuring that the toy withstands repeated use while providing an authentic racing experience. Manufactured primarily from PLA plastic, the car is lightweight yet sturdy, with rubber bands on the wheels to improve traction and performance. The propulsion system relies on a spring-loaded winding mechanism, allowing users to manually wind up the car and release it using a sliding pin to see it drive forward, demonstrating energy storage and conversion. To meet both user expectations and engineering constraints, the design had to balance aesthetics, material efficiency, ease of assembly and mechanical reliability. The scope of this project included SolidWorks modeling, material selection, prototyping and performance testing to refine the design for optimal functionality. By developing this toy, we aimed to create an engaging, sustainable alternative to battery powered toy cars while sparking interest in mechanical systems and engineering principles among young users.

1.2 Constraints & Criteria

Constraints are the fixed limitations that the toy must follow to meet the project requirements.

They are non-negotiable and must be satisfied for the design to be considered valid.

Table 1: Constraints

Constraint	Type	Description	Justification
Size	Assigned	All parts must fit inside a 10 cm tall egg, and when assembled it must be larger than the egg	Limits part size and ensures an actual assembly.
3D Printing Requirement	Assigned	75% of total toy weight must be 3D printed (PLA, ABS, PET).	Encourages use of modern fabrication methods.
Energy Storage and Release	Assigned	Must store potential energy and convert it to kinetic energy without user input after triggering.	Supports the use of 3D printing and SolidWorks.
No Pre-Stored Energy	Assigned	Toy must not arrive with pre-loaded energy.	Promoted safety.
Tool-Free, Adhesive-Free Assembly	Assigned	Must be assembled without tools, glue, tape, or melting etc.	Enhances child safety and reusability.
No Removable Pin	Assigned	Trigger mechanism must not use removable pins.	Avoids safety issues and promotes longevity.
No Commercial Components	Assigned	No use of toy parts from third party companies.	Only a collaboration between Kinder and our company.
Safety Requirements	Assigned	Toy must not be a safety risk or promote violence.	Safety is essential for child-friendly design.
Max Assembly Time	Team Defined	Must be assembled in under 2 minutes.	Improves usability and reusability.
Material Availability	Team Defined	Use only materials accessible via campus or standard suppliers.	Ensures feasibility.

Criteria are the standards used to evaluate how well the toy performs and how successful the design is. Unlike constraints, criteria are things we aim to maximize or improve to create the best possible solution.

Table 2: Criteria

Criteria	Type	Description	Justification
Functionality	Assigned	Toy must travel 2.5m as it is a running toy.	Main metric for successful performance.
Creativity and Innovation	Assigned	Unique design elements are rewarded.	Promotes originality.
Aesthetics	Assigned	Toy should look fun and engaging while aligning with sports theme	Important for user appeal and marketing.
Reusability	Team Defined	Toy should be resettable with minimal effort.	Encourages playability and longevity.
Durability	Team Defined	Toy must survive 10+ cycles of full use without mechanical failure.	Improves long-term value and user safety.

2 Design Process

The design process was thorough and lengthy. Initially, each group member created two sketches of possible toys to manufacture. Upon discussion, our team decided to blend various elements of each other's ideas together and settle on a soccer-themed projectile. Upon initial attempts to create the planned design, and after further deliberation, the team decided that a projectile toy would be much too difficult to make consistently accurate. The team quickly decided upon creating an F1-inspired race car — this idea would eventually become the “Blueprint 1.” At first, some team members were hesitant, believing car toys to be overdone and potentially boring. However, we recognized that with thoughtful design, strong aesthetics, and a well-executed energy mechanism, we could create a car that was both exciting and visually appealing.

To power the car, the team decided to create a spring and gear system. We designed a 3D-printed spiral spring to be twisted and tightened, storing potential energy. This spring was connected to a gear system that transferred the energy to the rear axle, causing it to spin. The stored energy was held in place by a sliding pin inserted into a hollow portion of a gear, preventing motion until the pin was slid out of the opening to release the energy.

After 3D printing, a few issues arose. The pieces were much too big to fit in the Kinder egg, and the wheels were too loose to hold onto the axles. We decided to reduce the size of the toy by almost half, as well as add a front and back wing for aesthetics. Upon reprinting, the toy now fit perfectly in the egg — however, issues continued to arise. The car did not meet the distance criteria and looked slightly bland. The team began writing down and brainstorming various fixes to the issues. We concluded that the spiral spring must be made larger, and more friction must be added to the rear tires.

The final print of the “Blueprint 1” contained a slightly larger spiral spring and more detailed wings. Additionally, elastic bands were wrapped around the rear tires. The new vehicle performed perfectly, excelling in all criteria. Throughout all of this, the team was experimenting with eye-catching colors. Our initial choice was a dark blue, selected to match the name “Blueprint 1” and emphasize the engineering-inspired theme of the toy. However, with further brainstorming, we decided to enhance the visual appeal by blending silver and blue. This combination maintained the original concept while giving the car a more sleek, high-performance appearance that better captured the excitement of motorsports and appealed to our target audience.

3 Design Alternatives & Evaluation

3.1 Conceptual Designs

Concept 1: Spring Powered Soccer Kicker

This Kinder Egg toy features a soccer player with a spring-loaded kicking mechanism. The toy comes disassembled for easy storage and assembly, with snap-fit joints requiring no adhesives. To play, the user pulls back a rod attached to the kicking leg, storing potential energy in the spring. When released, the leg swings forward, launching a foam ball. Adjustable joints

allow for aiming, and side supports help keep the ball in position before kicking. Some limitations include inconsistent launch distances, challenges in achieving precise aim, and potential wear on the spring or joints with repeated use, hence the final decision we came up with does not use springs as can be seen for Figure 4.

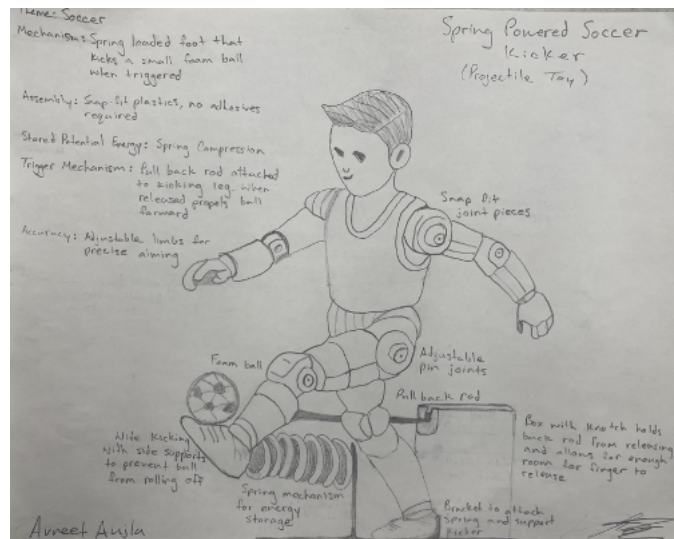


Figure 4: Concept Sketch #1- Spring. Soccer ball with spring locked leg that releases to shoot ball

Concept 2: Cricket Ball Thrower

This toy was our second concept for the projectile toy option within the Kinder Egg. To ensure the toy would be larger than the egg, yet it would still be able to fit within the Kinder Egg when closed, the toy was created to have pieces, so when bought, it will come disassembled with an instruction manual. When put together, a stand will be placed with a human-like athlete (with a pouch-like hand to hold the ball) on top of it. One of the arms of this toy will be able to be moved back and forth, with a fixed point placed on the stand on which an elastic band is wrapped around. This band is also wrapped around the cricket player's arm and pulled back to create potential energy, so that when released, the arm would slingshot and release the cricket ball, ensuring it crosses the 2.5-meter mark. The limitations we found in this concept was that the ball would not be able to cross the designated 2.5-meter mark, and ensuring the ball exits the hand at the correct time to get the proper elevation would become rather difficult.

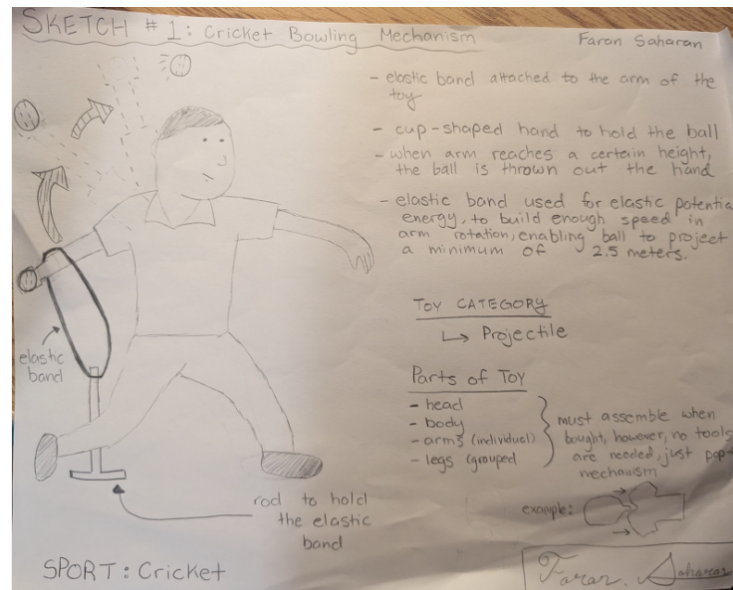


Figure 5: Concept Sketch #2- Cricket ball thrower. A cricket player placed on a stand that can be disassembled and fit into the Kinder Egg, with an arm that can be moved in a wind-mill motion pulled back via elastic and releases the ball.

Concept 3: Tennis Ball Launcher

This toy was our third concept, also for the projectile toy option within the Kinder Egg. To ensure the fully assembled toy would be larger than the egg while still fitting inside when disassembled, the design features multiple snap-fit pieces, including a two-piece racket and a two-piece base. When put together, the toy consists of a miniature tennis racket launcher mounted on a stand. A foam ball is placed inside the racket's frame and secured by a latch mechanism. The toy utilizes a spring-loaded system to store potential energy. When the push-and-slide trigger button is activated, the spring releases, launching the ball forward and simulating a real tennis shot. The adjustable base allows for aiming, ensuring the ball travels the required 2.5 meters. The limitations we found with this toy were that it may not be accurate. The face of the racket is so large, that when placing the ball on the racket, it could go in an undesired location, hence the accuracy would not be feasible enough to be accredited as a Kinder Egg toy. The other limitation is that the disassembling would only be of 2 parts, and the toy must be bigger than the egg when assembled, so it could be a challenge to figure out the sizing of each part of the toy.

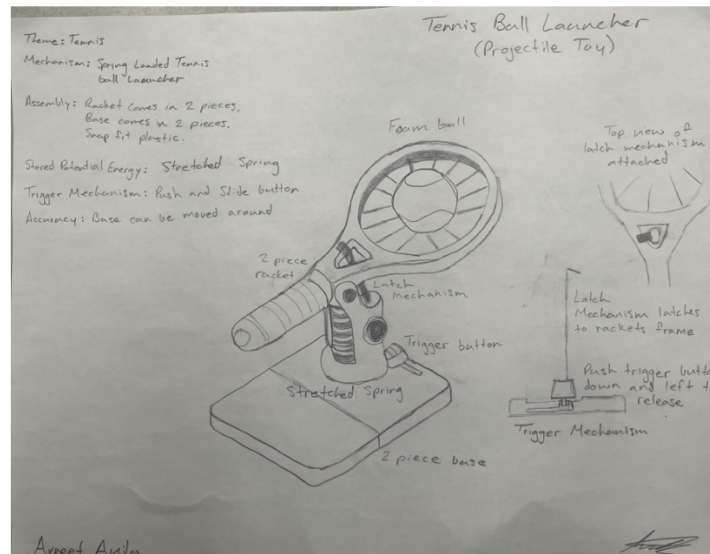


Figure 6: Concept Sketch #3 - Tennis Ball Launcher. A tennis racket launcher that can be disassembled and fit into the Kinder Egg, featuring a two-piece racket and base. The toy uses a spring-loaded mechanism to shoot a small foam ball off the racket.

3.2 Design Evaluation

To systematically evaluate and select the most suitable design for the Kinder Egg projectile toy, a structured assessment was conducted based on key engineering criteria, including energy storage efficiency, accuracy, ease of assembly, compactness and durability. Each concept was analyzed using qualitative rankings and a tradeoff approach to determine the optimal balance between functionality and practicality.

The spring powered soccer kicker (concept 1) effectively stores energy through spring compression and features adjustable joints for aiming. However, its accuracy can be inconsistent due to the unpredictable ball trajectory, and repeated use can lead to mechanical wear on the snap fit joints and spring. The cricket ball thrower (concept 2) relies on elastic potential energy to propel the ball but achieving a precise and repeatable launch is challenging. The hands grip on the ball and the release timing introduces variability, making it difficult to ensure the projectile meets the required 2.5-meter distance. Also, while the toy's design allows it to fit within the Kinder Egg, the reliance on an elastic band raises concerns about the long-term performance and wear over time. The tennis ball launcher (concept 3) offers the most reliable launch mechanism, utilizing a spring-loaded trigger system that provides consistent power delivery. The adjustable base improves aiming precision; however, the large racket face introduces variability in ball

placement, affecting shot accuracy. Ensuring the toy remains compact while maintaining structural integrity poses a design challenge.

While the cricket thrower and tennis ball launcher demonstrate engaging user interaction, they lack the optimal balance of energy efficiency and compactness. Despite concerns with minor accuracy inconsistencies, the soccer kicker is the best of the three options due to its effective energy storage, simple yet engaging mechanism reducing costs and ease of assembly. All three concepts are primarily 3D-printed, keeping material costs relatively similar. However, both the soccer kicker and tennis ball launcher incorporate small metal components, such as springs, axles and pins, which slightly increase their cost compared to the fully plastic cricket ball thrower. The soccer kicker's snap fit joining makes it easy to construct, and its spring-based propulsion delivers a reliable launch force while adhering to the Kinder Egg size constraints. Accuracy is prioritized more heavily; a guided ball placement or a constrained kicking path can be considered. However, this would drive up the cost due to additional material requirements. Alternatively, compactness is also a primary concern and a further reduction in part count can reduce costs. It's important to find a balance between guides and part count as launcher accuracy as well as being cost effective are important factors in making a consistent yet cost effective launch toy. Finding a balance between guided accuracy and minimal part count is essential to making a consistent yet cost effective launch toy. A final design that tackles all these issues while prioritizing enjoyment must be found.

Table 3: Decision Matrix for Concept Sketches. The three possible initial options are tested and evaluated based on five categories

Criteria (/5)	Criteria Weight (%)	Soccer Kicker	Cricket Ball Thrower	Tennis Ball Launcher
	Rating (/5)			
Aesthetics	20	5	2	3
Ease of Modeling	15	2	4	4
Mechanism Function	25	4	3	3
Cost	25	4	5	4
Size	15	3	3	2
Weighted Total		75	69	65

4 Proposed Design

Table 4: Decision Matrix for Toy Selection. The wind-up racecar was selected for further development, primarily due to its high mechanism function and ease of modeling.

Criteria (/5)	Criteria Weight (%)	Wind-up Car
	Rating (/5)	
Aesthetics	20	4
Ease of Modeling	15	5
Mechanism Function	25	5
Cost	25	4
Size	15	4
Weighted Total	100	88

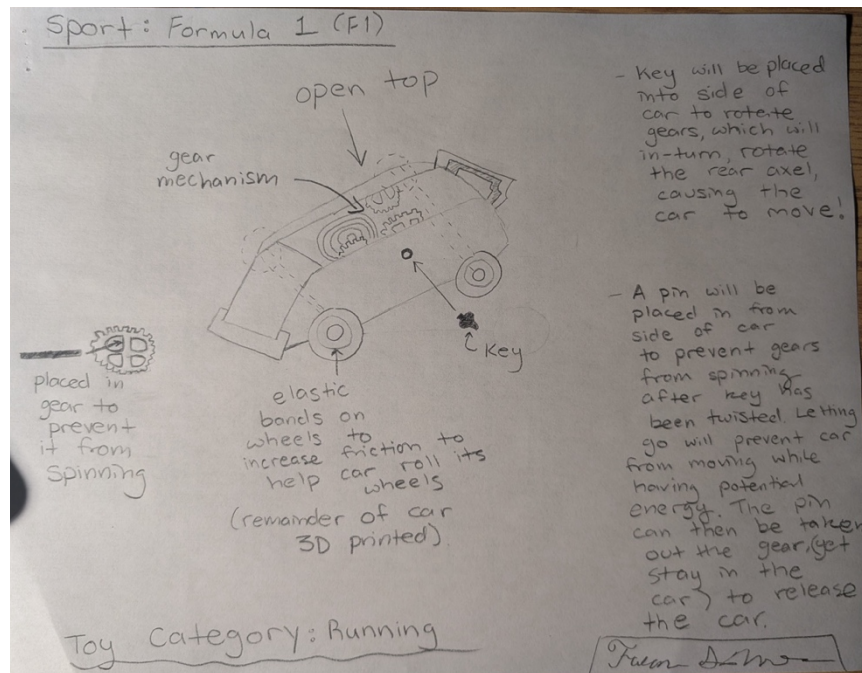


Figure 7: Chosen Toy Concept Sketch – F1 Car. A toy car that has F1 features and has a wind-up mechanism

4.1 Design Description

The design selection process for the wind-up toy race car involved a structured and objective evaluation of various potential design solutions, ensuring that the final product met performance, manufacturability and user experience requirements. The primary focus was on optimizing the propulsion system, chassis structure and wheel design to balance durability and ease of use. A combination of research, SolidWorks modeling, prototyping and iterative testing was used to evaluate different configurations and refine the final design.

The key functional component of the toy car is the spring driven wind-up mechanism as seen in Figure ____ as well as the 3D SolidWorks pieces in Appendix A, which stores and releases mechanical energy to propel the car forward. During development, multiple spring (Appendix A) sizes were tested to determine the optimal balance between winding effort, energy storage capacity and release control. Smaller springs required less force to wind but stored insufficient energy, resulting in shorter travel distances. In contrast, larger and stiffer springs provided greater energy storage but required excessive force to wind, making the toy difficult for children to use. After iterative testing, a medium stiffness spring was selected, as it provided an ideal balance between user friendly operation and effective propulsion.

The chassis (Appendix A) was designed to be both lightweight and durable while ensuring structural integrity. Initial prototypes featured solid PLA construction, however testing revealed that this resulted in unnecessary weight, reducing acceleration and overall performance. To resolve this, a revised design incorporated internal lattice structures and shelling some parts, which significantly reduced weight while maintaining sufficient rigidity. PLA plastic was chosen for its ease of 3D printing, affordability and environmentally friendly properties compared to other thermoplastics such as ABS.

Wheel (Appendix A) performance played a crucial role in determining the overall effectiveness of the design. Early prototypes used fully 3D-printed PLA wheels, which suffered from low traction, causing inconsistent movement, particularly on smooth surfaces. To improve grip and stability, rubber bands were added around the wheels. This modification significantly enhanced traction, allowing the car to travel further and more consistently. While the addition of rubber bands increased manufacturing complexity, the improvement in performance justified the tradeoff.

Throughout the evaluation process, design decisions were made based on key criteria, including durability, ease of assembly, performance consistency and user interaction. If different weightings had been applied to these criteria, alternative design choices might have been selected. For example, prioritizing cost reduction over performance might have led to a simpler, weaker spring mechanism, while emphasizing aesthetics over functionality could have resulted in a heavier, less efficient chassis. However, given the project's goal of creating an engaging and educational toy, the focus remained on optimizing durability and mechanical efficiency.

The final design successfully balances performance, durability and manufacturability. The selected spring mechanism provides reliable and repeatable motion, the PLA parts maintain strength while minimizing weight, and the rubber enhanced wheels ensure stable movement. While certain tradeoffs were necessary, such as increased complexity in wheel manufacturing versus improved traction, the overall design effectively meets the intended criteria. Future iterations could explore alternative spring configurations or materials to further enhance efficiency while maintaining ease of production. By employing a systematic and iterative evaluation process, Blueprint 1 was developed as a functional and engaging toy that demonstrates fundamental engineering principles while offering a practical and sustainable alternative to battery powered models.

4.2 Assembly Instructions

The assembly process for Blueprint 1 is designed to be simple and intuitive, requiring no tools. The components are designed to snap together securely, ensuring ease of assembly while maintaining durability and functionality. The entire process consists of seven steps, carefully structured to ensure proper alignment and performance of all mechanical and structural elements. The assembly begins with the right body, which serves as the foundation for the car. The cross members are first snapped into place, providing structural support. The right pinion guide is then secured, and the spring axle is carefully slid through its designated hole in the right body. Once in place, the front and rear axles are inserted into their respective slots, ensuring they rotate smoothly with minimal friction. Next, the spring is fitted onto the spring axle and idler, ensuring proper engagement with the drivetrain. Following this, the remaining gears are carefully positioned to complete the mechanical assembly allowing for efficient transfer of stored energy from the spring to the wheels. The left pinion guide and idler spacer are then placed into position

to ensure proper alignment and smooth operation of the gear system. With the internal mechanism assembled, the left body is snapped into place, enclosing the drivetrain and securing all internal components. The wheels are then pressed onto the axles, ensuring a firm fit. Finally, the front and rear wings are attached by fitting them onto the cross members, completing the aerodynamic and aesthetic aspects of the design. To verify assembly, users should test the winding mechanism and ensure the wheels rotate smoothly when released. For a more detailed step by step guide, including diagrams, please refer to the instruction sheet in Appendix C.

4.3 Manufacturing & Cost Analysis

For the prototype's production, the components were 3D printed with PLA. To allow the car to function and make it look more aesthetically pleasing, the aids of elastic bands and paint were bought from Amazon and Michael's. A summary of all materials and expenses are outlined:

Table 5: Cost of singular toy production

MATERIAL	MASS (g)	COST/UNIT
Material Print (PLA) [1]	70.05 g	\$1.40
Rubber Bands [2]	2.66 g	\$0.52
Paint [3]	15 g	\$0.38

Production Cost/Toy: \$2.30

Table 6: Annual cost and expenses for Blueprint Corporation

EXPENSES	COST
COST OF 42,000 TOYS	\$96,600
FACTORY COST [4]	\$20,000
PACKAGING/SHIPPING COST [5]	\$84,000
LABOUR COST	\$40,000

INJECTION MOULDING MACHINE [6]	\$15,000
TOTAL EXPENSES	\$255,660
TOTAL PROFIT	\$282,489
KINDER PROFIT ALLOCATION (15%)	\$49,851

Sale/toy: \$14.00

Total Revenue: \$588,000

To design the prototype, a 3D printer was utilized to finalize the product. For mass production to be cheaper and run smoothly, an injection molding machine is incorporated replacing a 3D printer. First, to be able to manufacture 42000 toys/year, a 3D printer is not optimal as it is not used to manufacture large quantities of components, Second, the cost of a 3D printer will be a lot more expensive annually as it has a higher cost per unit at scale. Lastly, not only does it have better material strength, but it also has a much more consistent surface finish.

To improve mass production, a thorough analysis was conducted in the production of the toy. To reduce energy and scrap, therefore lowering costs, we would optimize the injection molding machine. This can be done by reducing cycle time, regular maintenance, and using energy-efficient heaters.

There are a couple of constraints in this process. First, the packaging cost may increase as Kinder would like a specific process for the chocolate to be firmly done around the egg. Next, there is a huge risk with investing in an injection molding machine to set up a whole production line, as it may not be necessary for short term production.

5 Design Defense

Upon careful consideration, our team decided to make an F1 inspired race car, the "Blueprint 1". This decision was based on its reliable performance, compliance with all project constraints, and strong alignment with the "Sports" theme. Our original concept, a soccer themed projectile toy, was ultimately ruled out due to the difficulty in meeting the required accuracy.

Similarly, we chose not to pursue a floating or time-based toy, as both concepts introduced significant challenges. Floating mechanisms presented stability and space issues within the Kinder egg, while time-based toys required complex gearing or continuous motion systems that were difficult to achieve. In contrast, a running toy offered the most straightforward path to consistent functionality. Its wind-up mechanism efficiently stores and releases potential energy, allowing the toy to consistently travel over 2.5 meters and meet the functional requirement for “running” toys. The exterior design was carefully curated, the shiny blue metallic colour was specifically selected to attract our target audience. The front and back wings were also strategically designed to both attract attention to our product and ensure the toy would be larger than the kinder egg when assembled. The top of the Blueprint 1 race car was purposely exposed so that children would be able to view the spiral spring and gears working together to produce movement. The toy was also designed for simple, tool-free assembly, using no adhesives, ensuring it remains accessible and safe for children. By meeting all design constraints and maximizing key criteria like functionality, aesthetics, and ease of use, Blueprint 1 offers a dynamic, engaging, and well-engineered solution that captures the spirit of Kinder and the thrill of motorsports.

6 Marketing Strategy

Blueprint 1 is designed for children aged 7 and up, as there are small pieces that can prove as a choking hazard under this age limit. The toy features a colorful design, a simple assembly process without tools, and a fun winding mechanism that teaches basic physics concepts. To prevent health hazards, a clear safety warning is included in the instruction manual. This aligns with the goal of making an interactive yet safe toy for young individuals.

The target market consists of children between ages 7 to 12, as well as parents and guardians who make purchasing decisions. This demographic values toys that are fun, interactive, and educational. Additionally, parents seek safe and durable toys that provide long-lasting entertainment. Geographically, the initial focus is on Canada, particularly in urban and suburban regions where families frequently visit toy stores and engage in online shopping. Future expansion into international markets will be considered based on demand and distribution opportunities.

To effectively reach the audience, a digital marketing strategy centered on Instagram (Appendix D) has been implemented, a platform widely used by both parents and children. The content strategy includes bright visuals, engaging animations, and modern music, making the toy relatable to kids while also appealing to parents. Branding embraces playful energy and excitement, aligning with contemporary children's entertainment trends. Additionally, toy review platforms, parenting blogs, and digital advertisements will be leveraged to expand reach and credibility.

The pricing strategy considers material, manufacturing, and marketing costs, ensuring affordability without compromising quality. The toy is made from durable plastic, with an efficient production process that minimizes costs while maintaining a high-quality finish. By balancing affordability with durability, the product remains competitive within the toy market. Furthermore, design elements, branding materials, and marketing content constitute intellectual property that distinguishes the product from competitors.

By integrating a strong marketing approach, an age-appropriate design, and cost-effective production, the wind-up toy car offers an engaging, safe, and educational experience for children. With strategic branding and a focus on digital outreach, the product stands out as an exciting choice for both kids and adults.

7 Project Management

Effective project management was essential to ensuring that the toy car was developed efficiently within the given timeline and budget. The team followed a structured approach, which included careful scheduling, cost management, and overcoming design challenges. This section outlines the timeline, budget and problem-solving strategies used during the project.

Schedule

The project was executed in phases, beginning with initial design brainstorming and concept sketches. The team collaborated on developing an early prototype, which was tested and refined before moving to the final product. SolidWorks modeling was completed early in the process to ensure accurate dimensions and ability to 3D print. The first prototype was produced and tested for functionality, and adjustments were made based on fit and assembly issues. The final product

was then printed, painted and assembled, followed by final testing to ensure all components worked as intended.

Budget

The budget for the prototype was primarily allocated to 3D printing material, rubber bands for functionality and paint for aesthetics. The total weight of the 3D printed components was 70.05 grams, using PLA filament at a cost of \$0.07 per gram, bringing the total 3D printing cost to \$4.90. The design required 12 rubber bands, each costing \$0.04, totaling \$0.48 for the rubber bands. Paint was used to enhance the appearance, costing \$0.38. Before reaching the final design, an initial prototype was produced to test the design and assembly. The early prototype had a slightly different weight and required additional filament, resulting in a total cost of \$6.20. Including both prototypes, the total budget for the development phase was \$11.96.

Table 7: Total Budget for Development Phase of Kinder™ Egg Toy

Category	Cost
PLA Filament (70.05 grams) [1]	\$4.90
Rubber Bands (12) [2]	\$0.48
Paint [3]	\$0.38
First Prototype (Test Print)	\$6.20
Total Design Cost	\$11.96

Challenges and Mitigations

During the first test print, it was discovered that the sizing of the pieces was too large to fit inside the Kinder™ egg. This required a redesign of several components to ensure they could be properly assembled and stored while maintaining the toy car's functionality. Additionally, the second first complete prototype of the toy car did not travel the intended 2.5-meter distance, prompting the team to modify the spring mechanism by increasing its size to store and release more energy effectively. This redesign improved the car's performance while keeping it within the Kinder™ egg's constraints. Beyond design challenges, the team initially faced scheduling difficulties in coordinating meetings due to conflicting availabilities. To resolve this, a shared calendar was introduced, where each member could input their availability. This approach helped

streamline communication and allowed the team to find common meeting times efficiently, leading to a more structured workflow ensuring the project was completed on time.

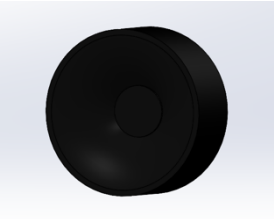
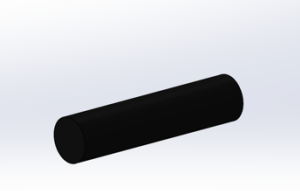
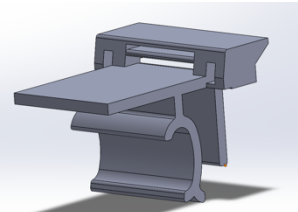
8 Conclusion & Recommendations

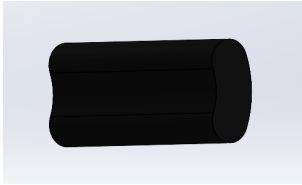
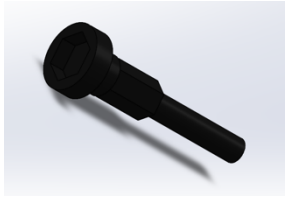
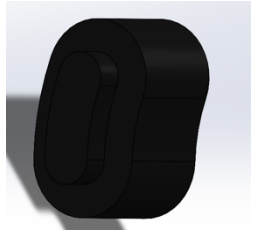
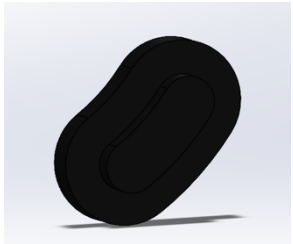
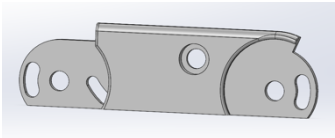
Our team at Blueprint 1 came up with the idea of a toy car powered by mechanical wind-up energy to create a fun, interactive experience for children without relying on disposable batteries. This approach not only makes the toy more environmentally friendly but also encourages kids to engage with simple mechanics and understand how the car moves through play. Early in the design process, we explored various concepts including projectile-based toys but ultimately decided to rule them out. Not only do they pose potential safety hazards for children, but they also come with design challenges, such as ensuring consistent accuracy and functionality. Given these concerns, we focused on a design that balances safety, creativity, and hands-on play, which led us to the wind-up car concept. Looking ahead, there's also potential to expand the concept by developing simpler versions of the toy geared toward younger children, allowing us to reach a broader audience while staying true to our focus on safe, creative, and sustainable play.

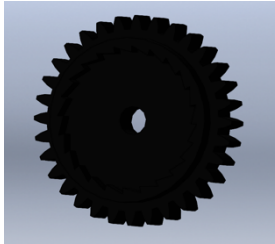
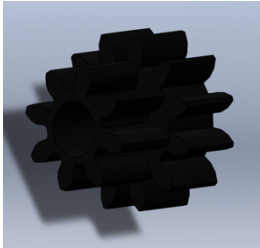
9 References

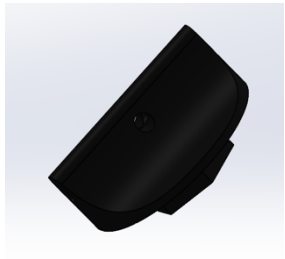
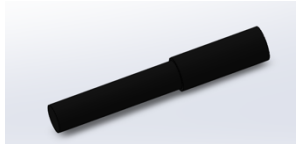
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Appendix A

Part Name	# of Prints	Part Image
Wheel	4	
Pinion	1	
Spacer	1	
Front Wing	1	
Back Wing	1	

Cross Member	2	
Front Axel	1	
Rear Axel	1	
Spring Axel	1	
Right Pinion	1	
Left Pinion	1	
Right Body	1	

Left Body	1	
Gear 1	1	
Gear 2	1	
Gear Sprawl	1	
Gear 12-8	1	

Pawl	1	
Key	1	
Idler	1	
Spiral	1	

Appendix B

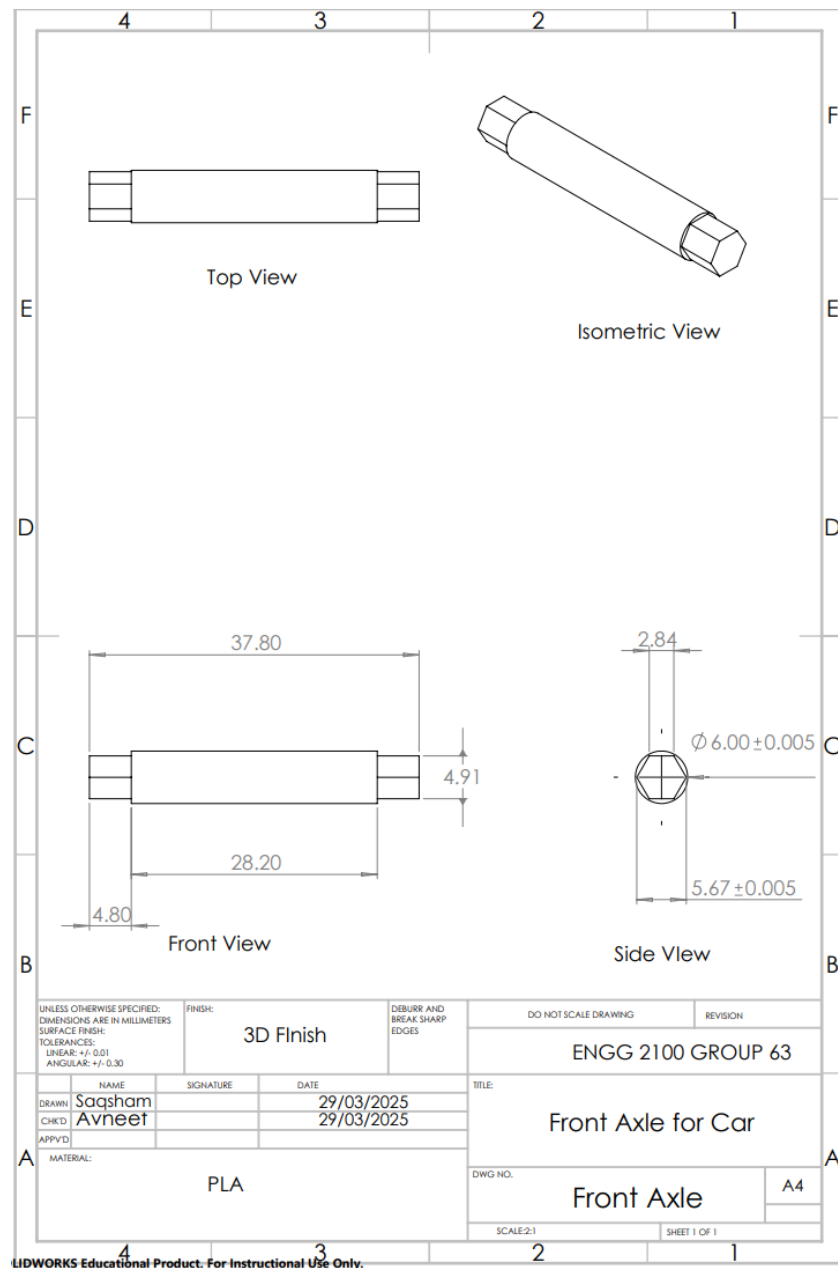


Figure 8: 2D Drawing - Front Axel

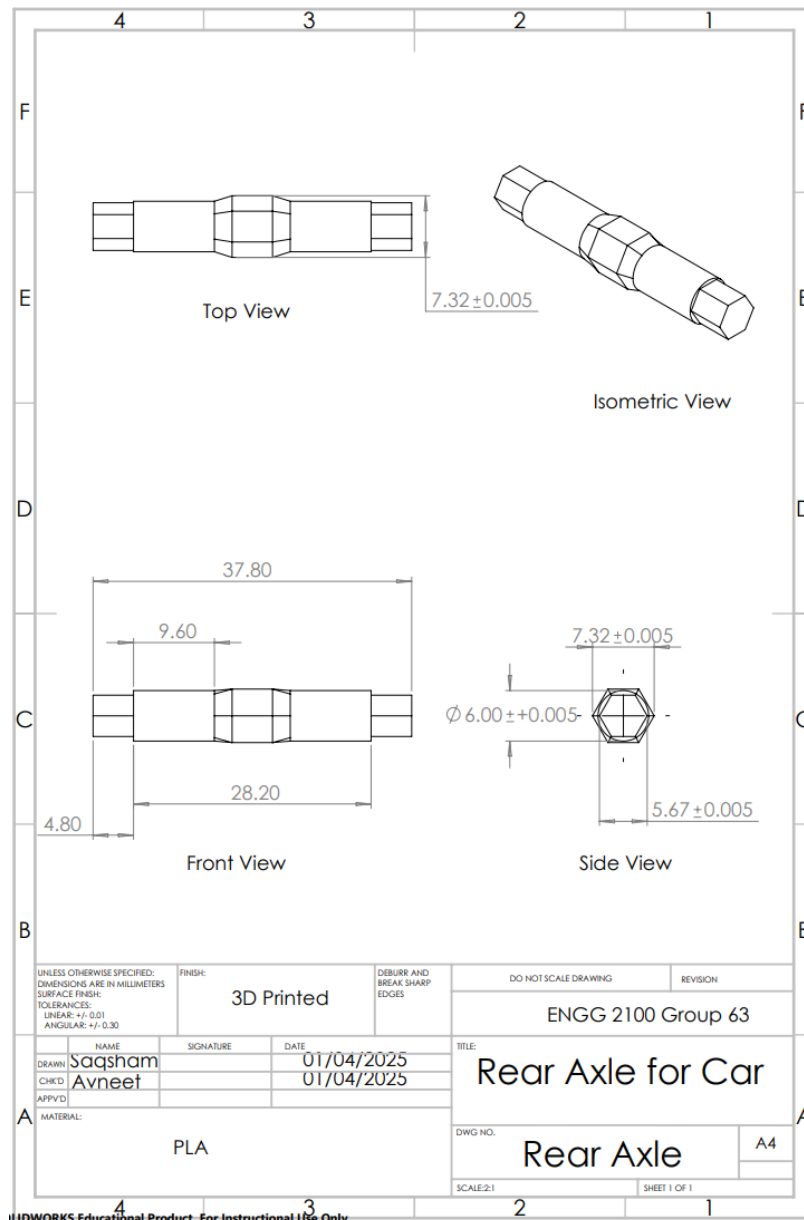


Figure 9: 2D Drawing – Rear Axel

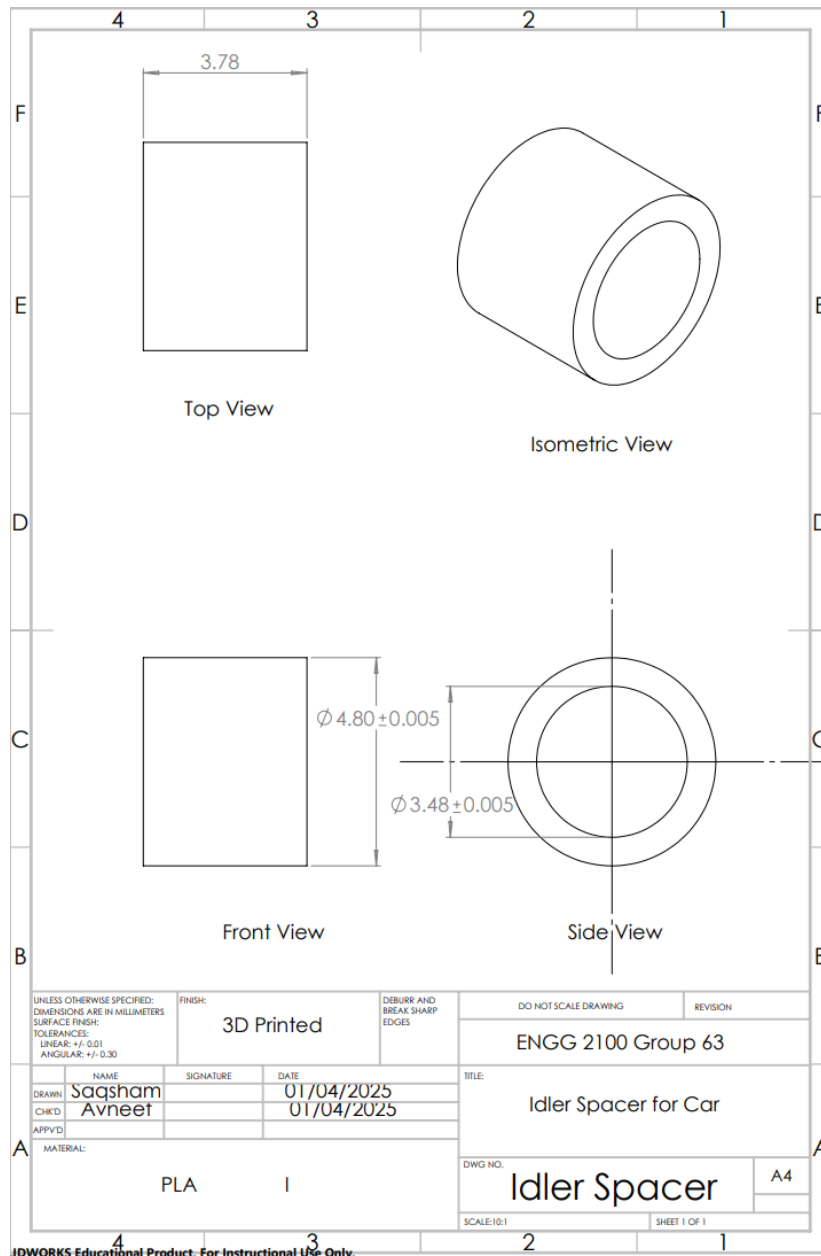


Figure 7: 2D Drawing – Idler Spacer

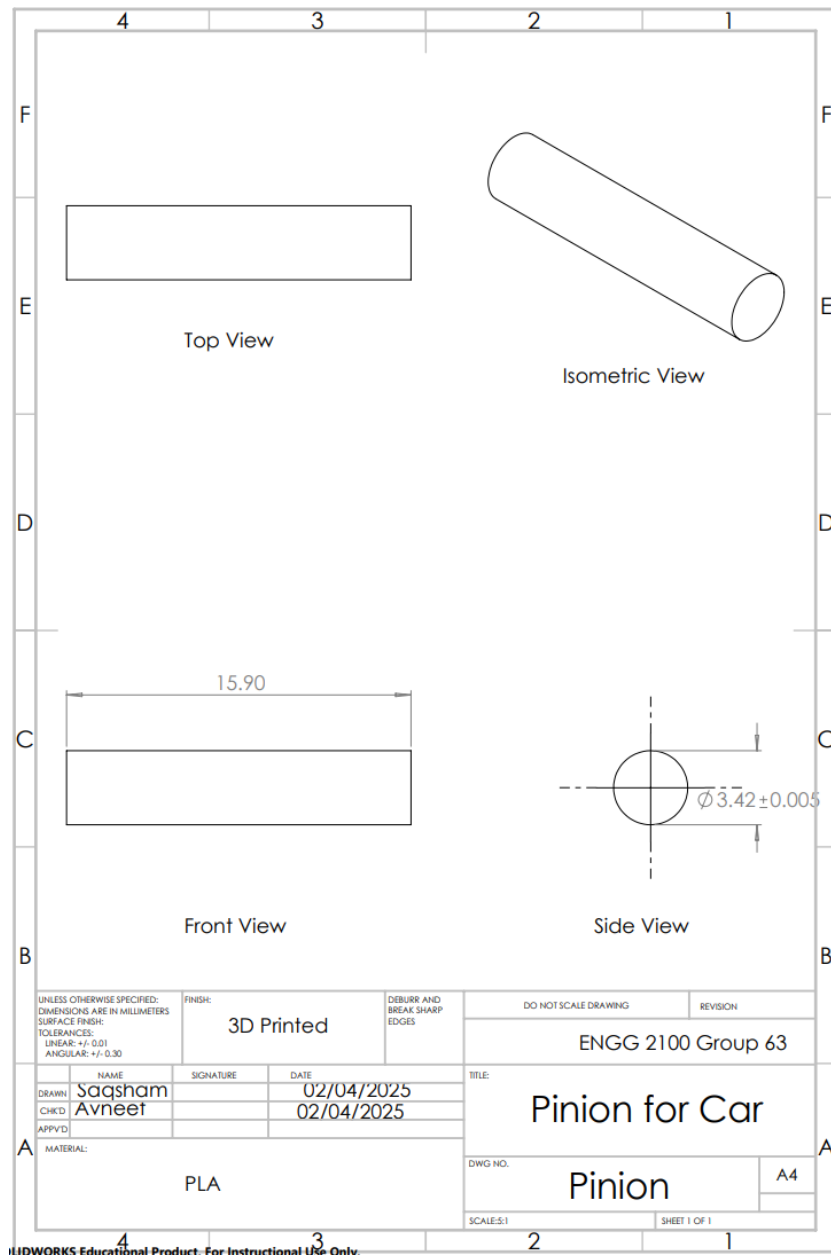


Figure 8: 2D Drawing – Pinion

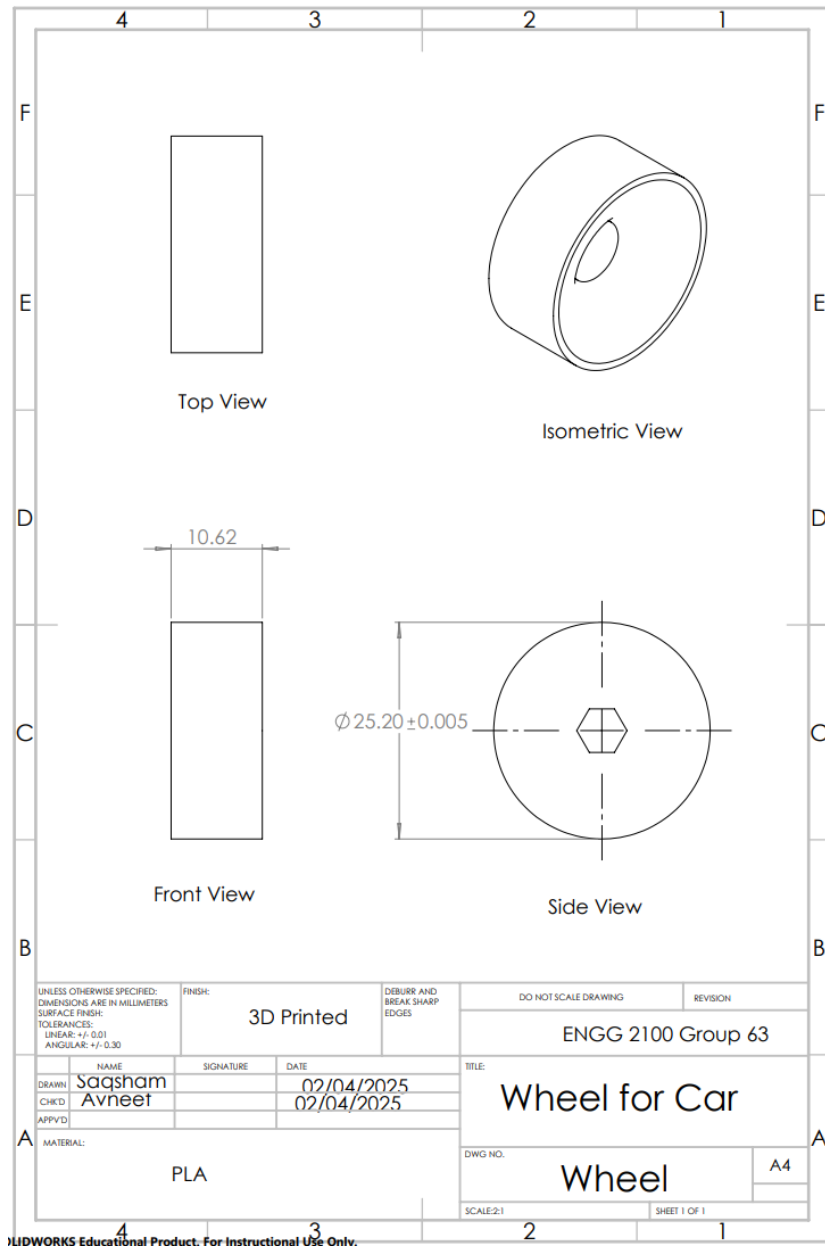


Figure 9: 2D Drawing – Wheel

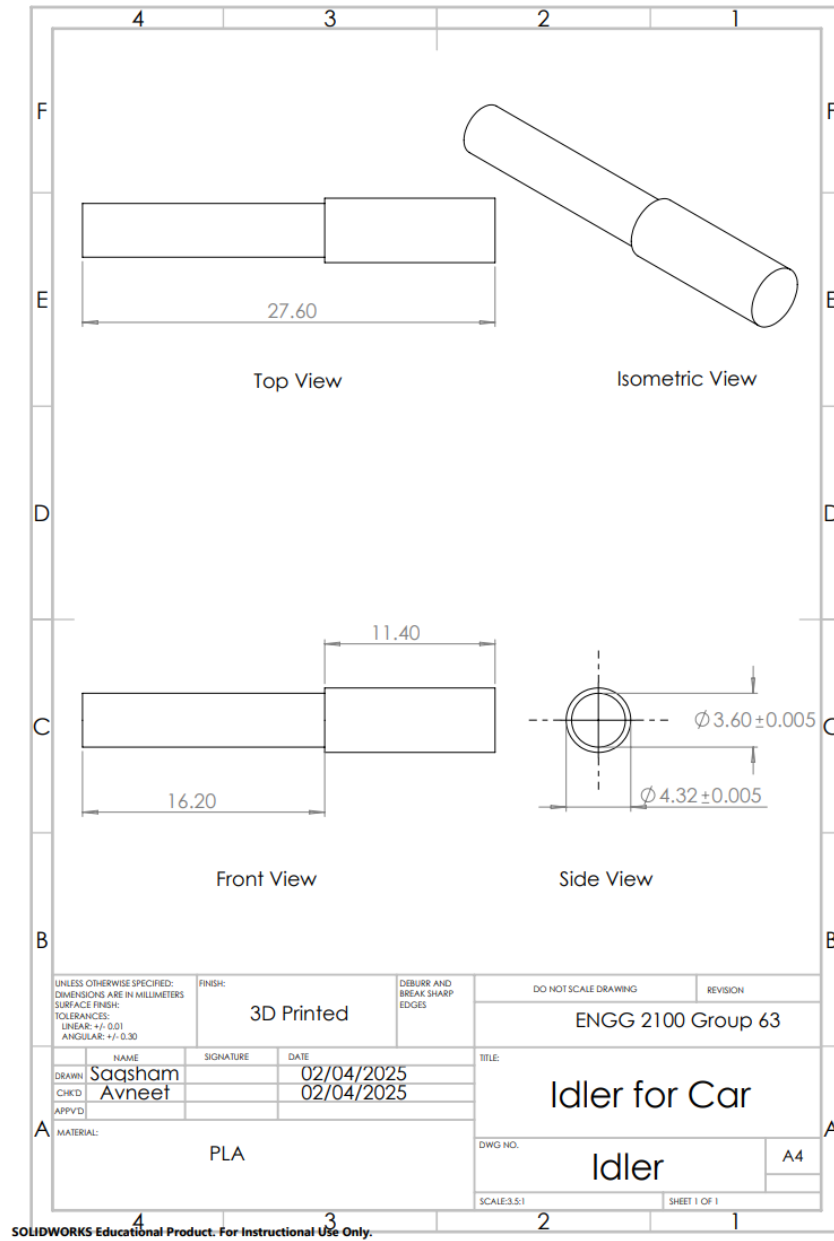


Figure 10: 2D Drawing – Idler

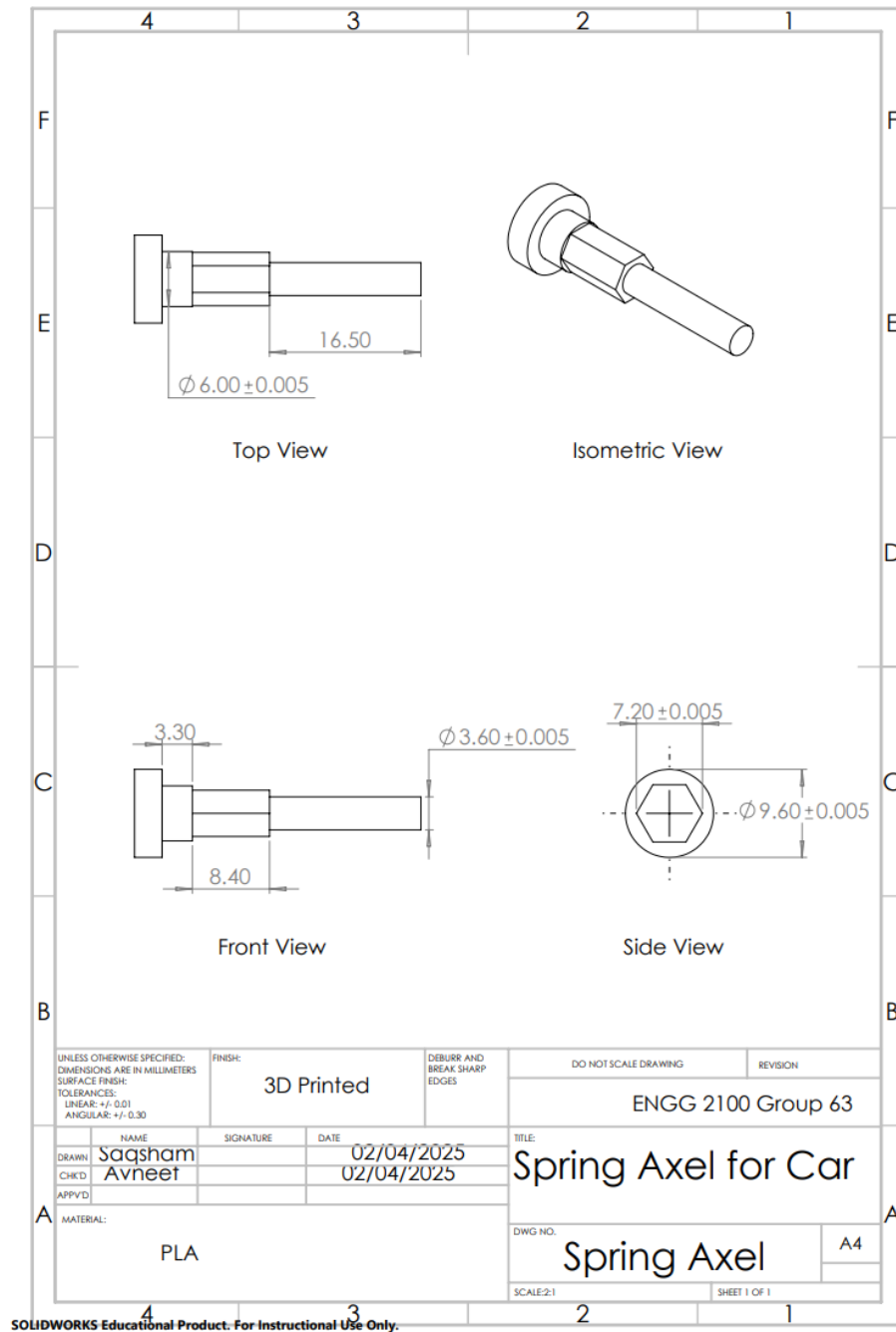


Figure 11: 2D Drawing – Spring Axel

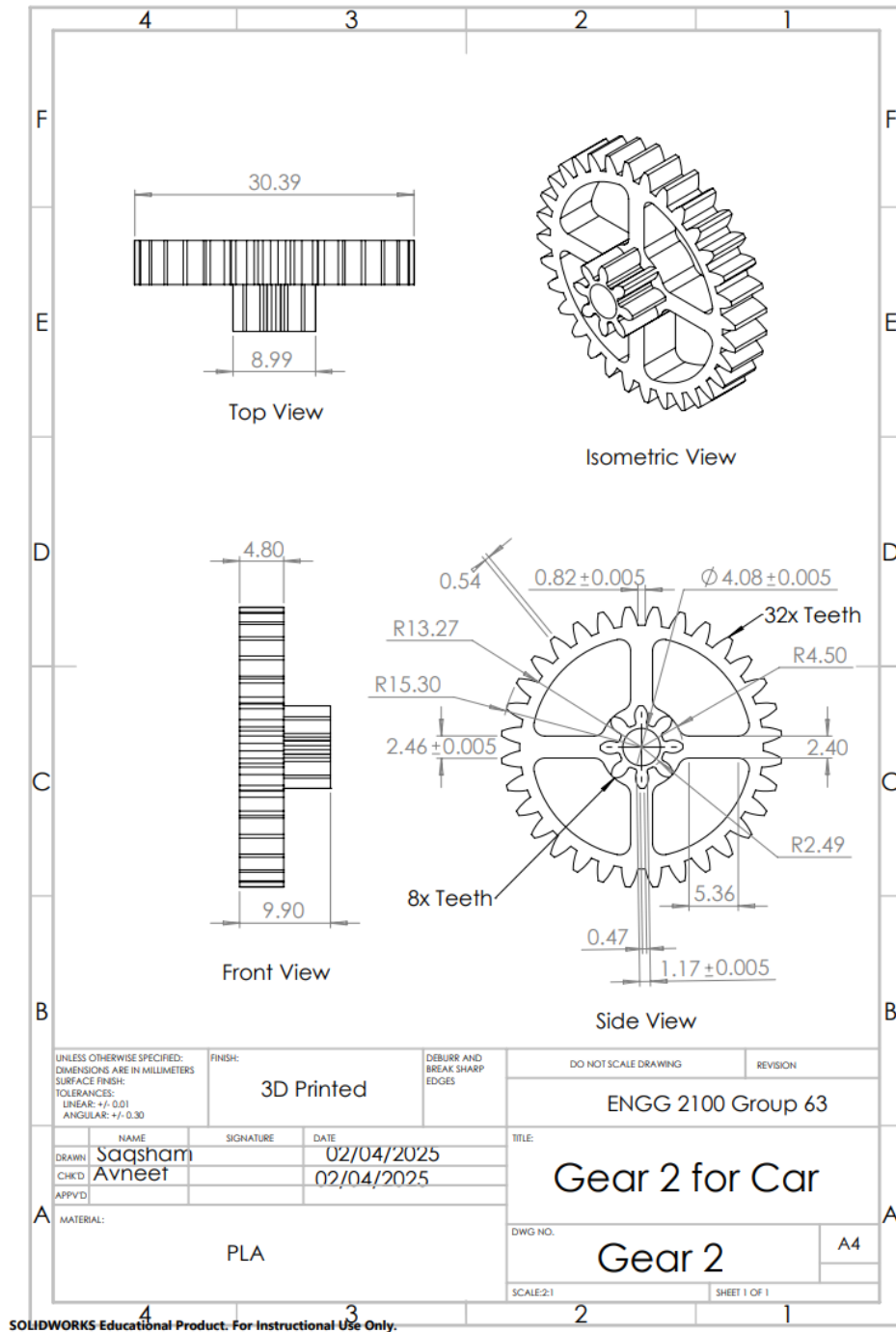


Figure 12: 2D Drawing – Gear 2

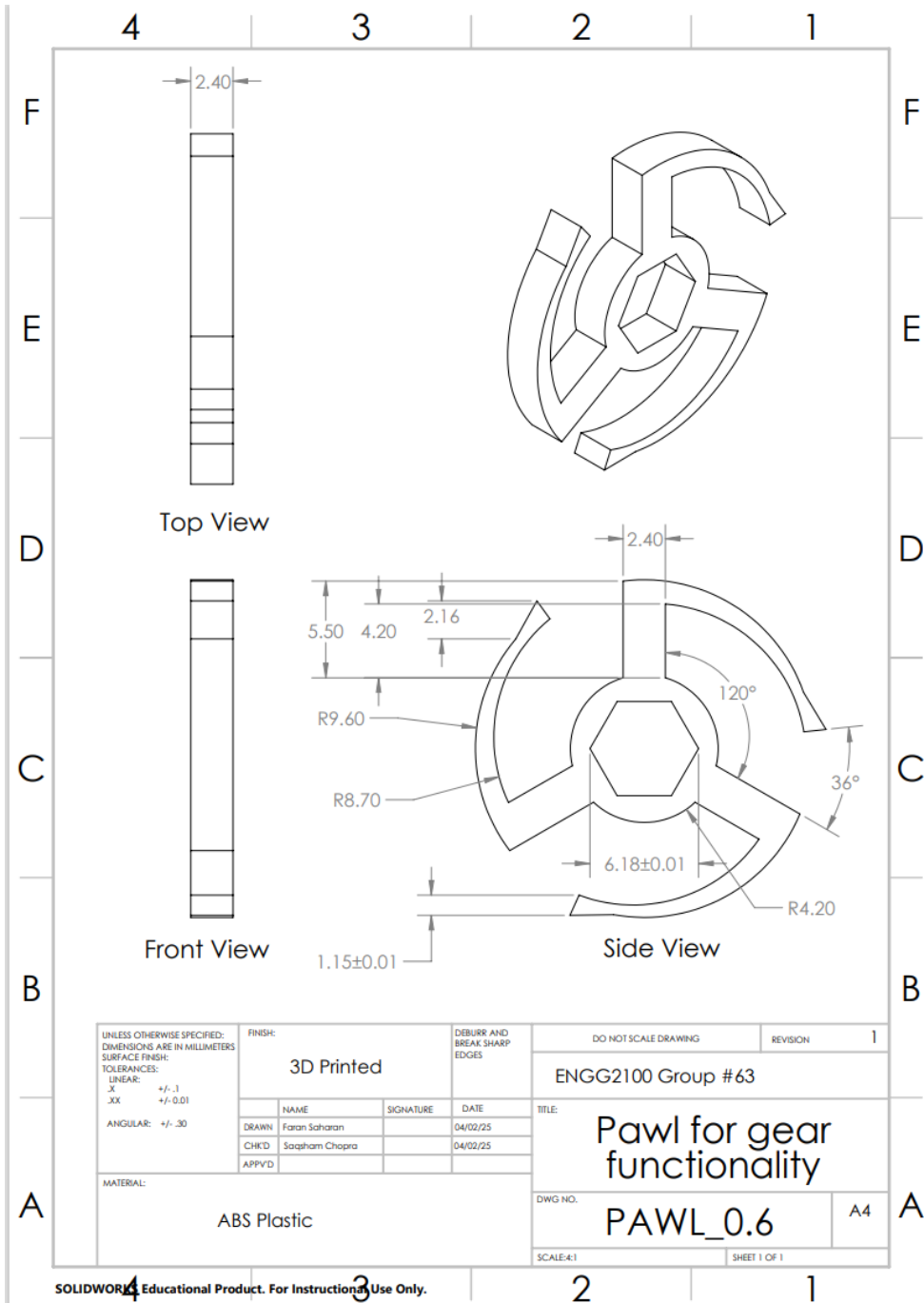


Figure 13: 2D Drawing – Pawl

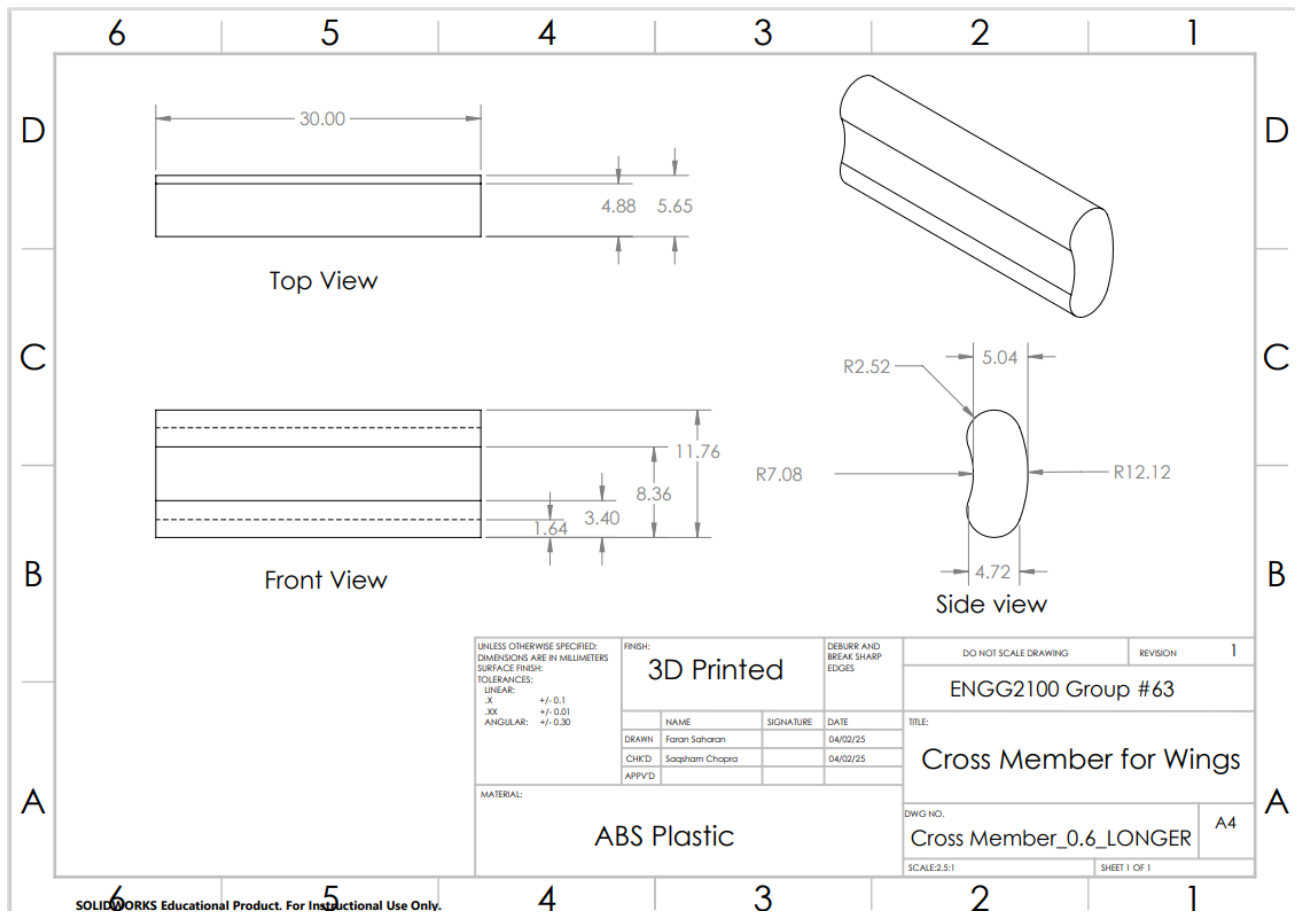


Figure 14: 2D Drawing – Cross Member

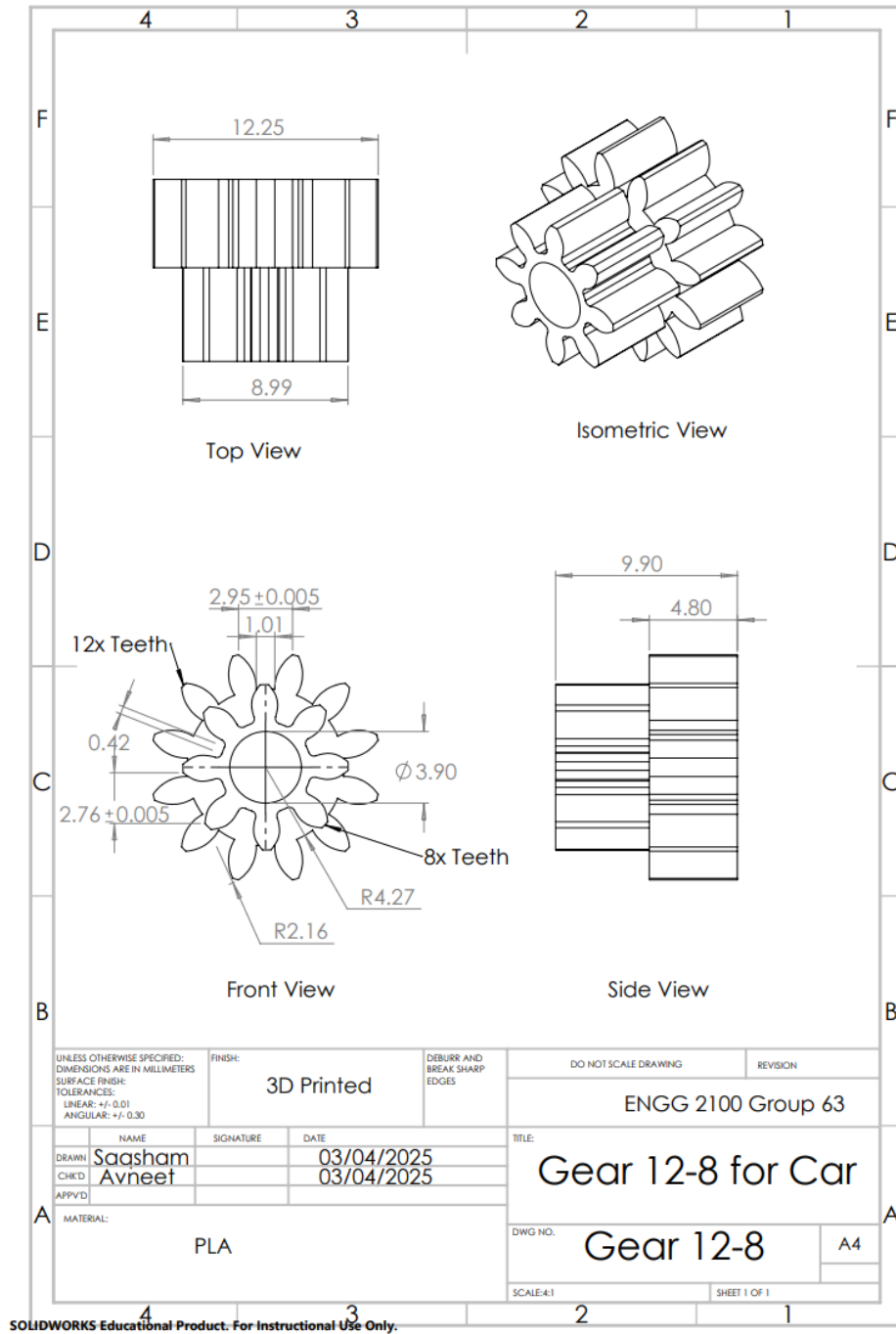


Figure 15: 2D Drawing – Gear 12-8

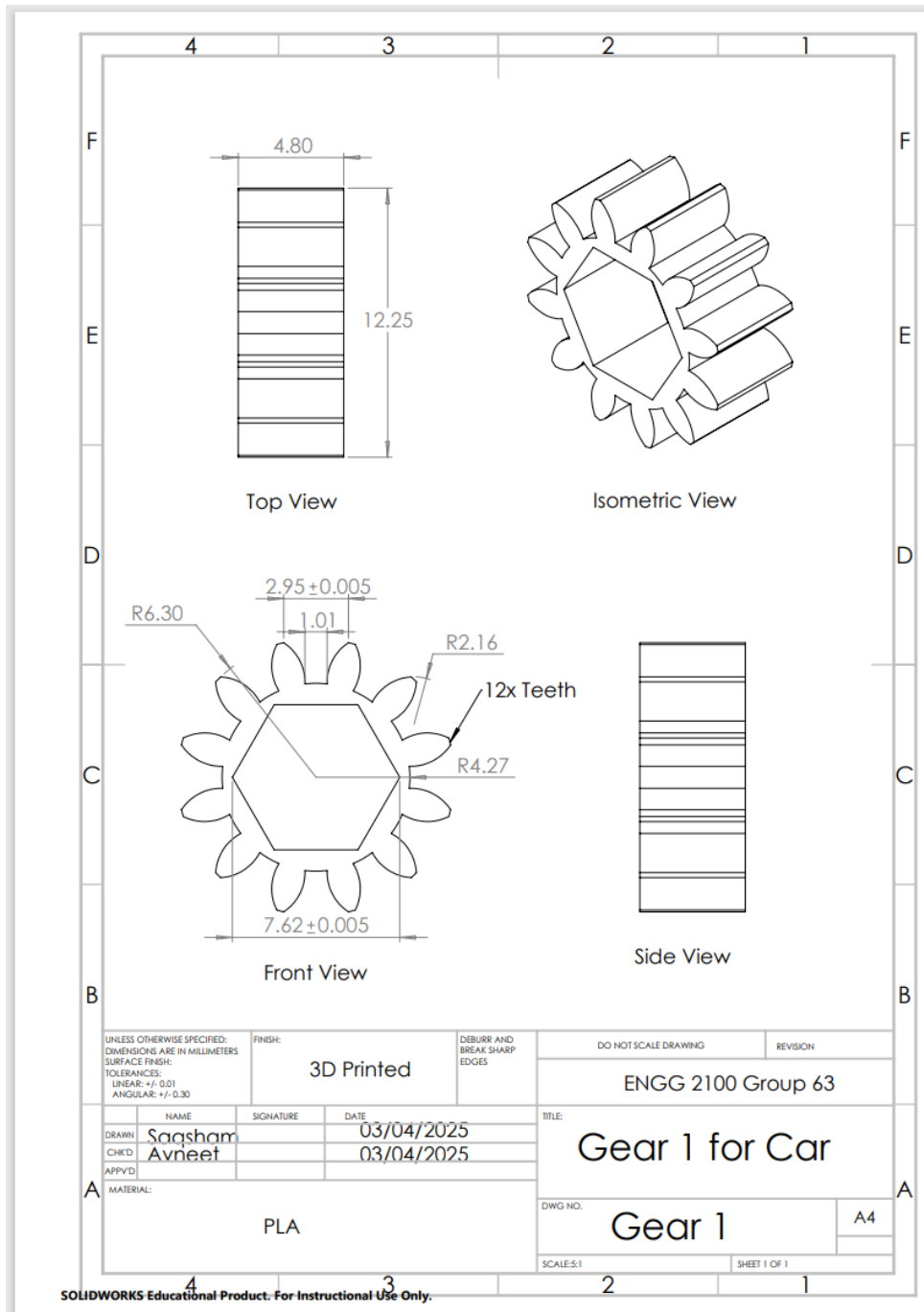


Figure 16: 2D Drawing – Gear 1

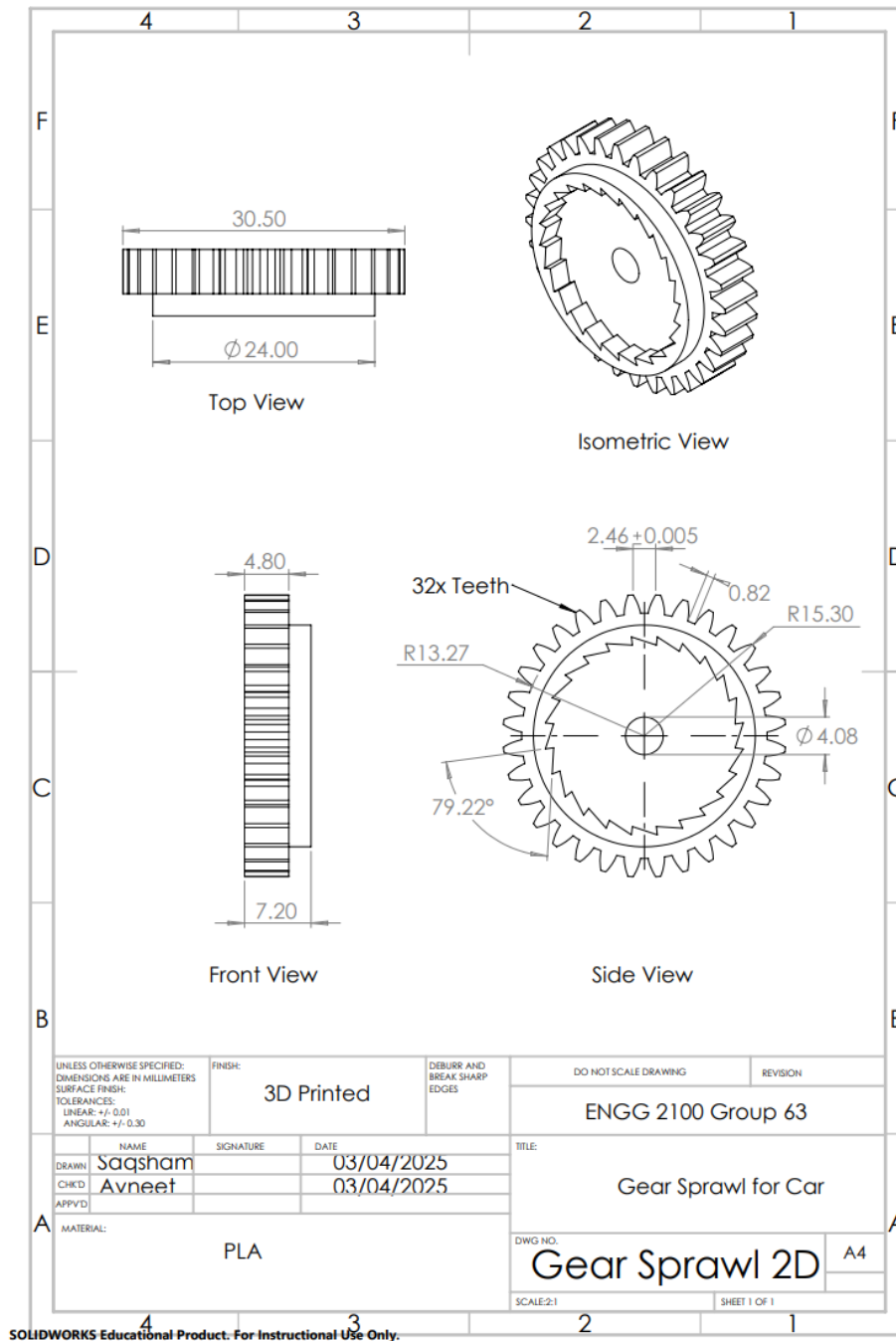


Figure 17: 2D Drawing – Sprawl Gear

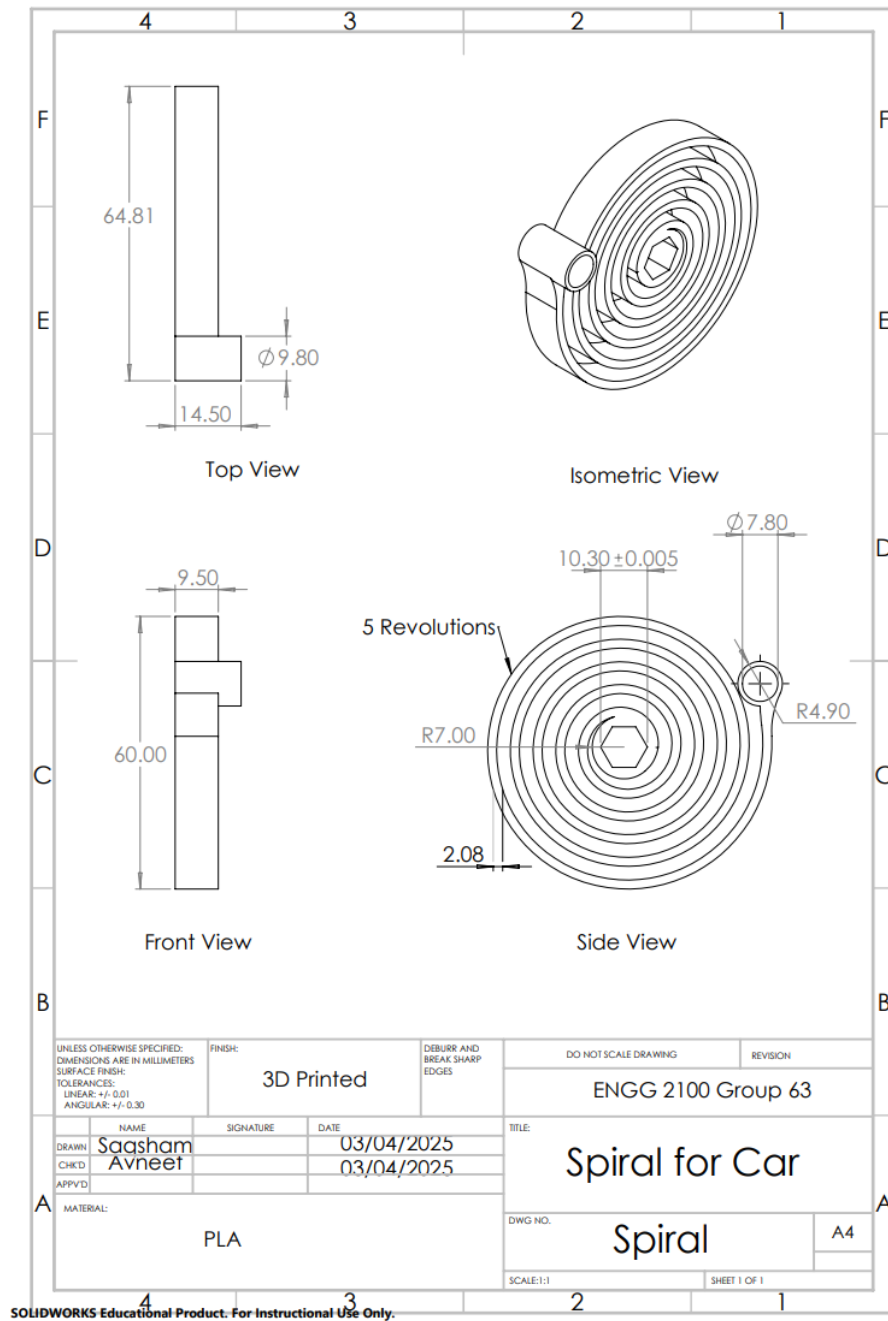


Figure 18: 2D Drawing – Spiral

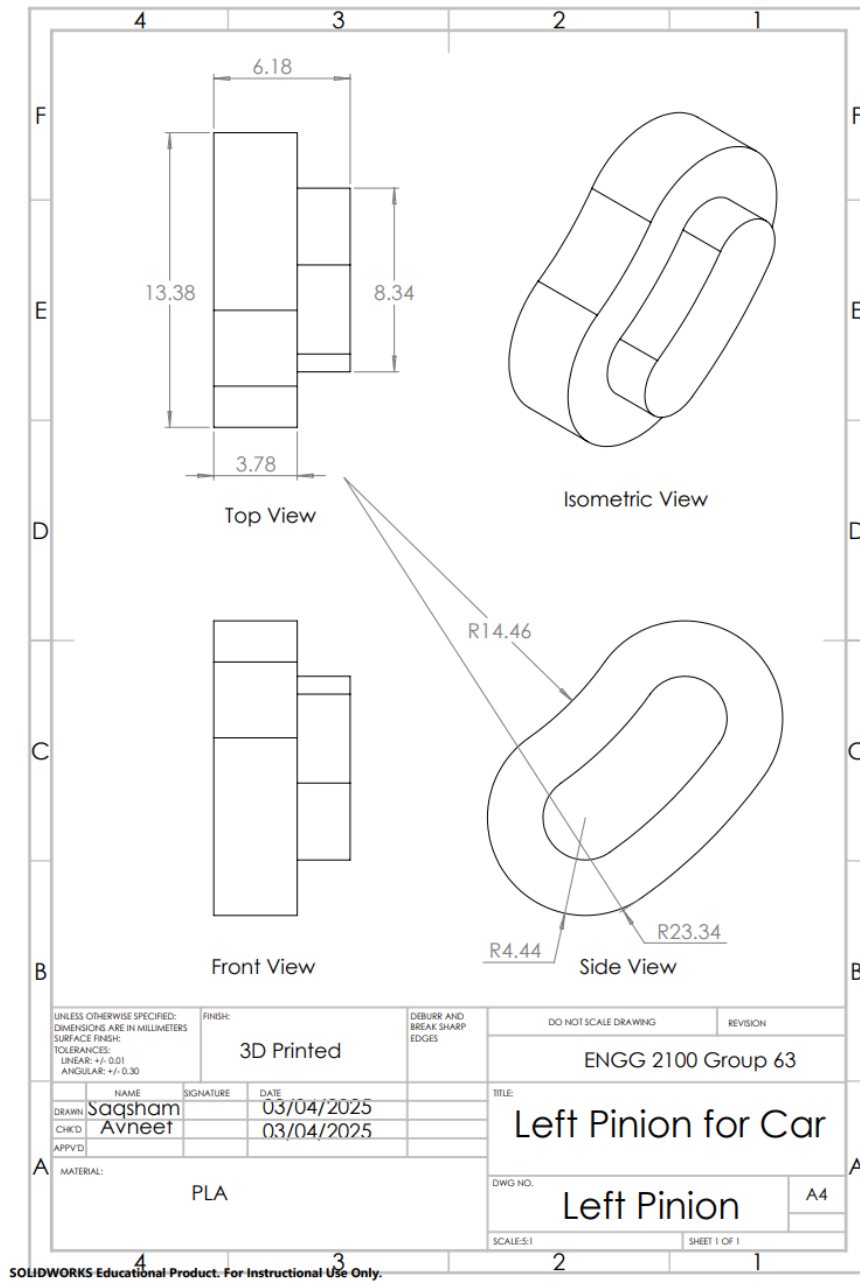


Figure 19: 2D Drawing – Left Pinion

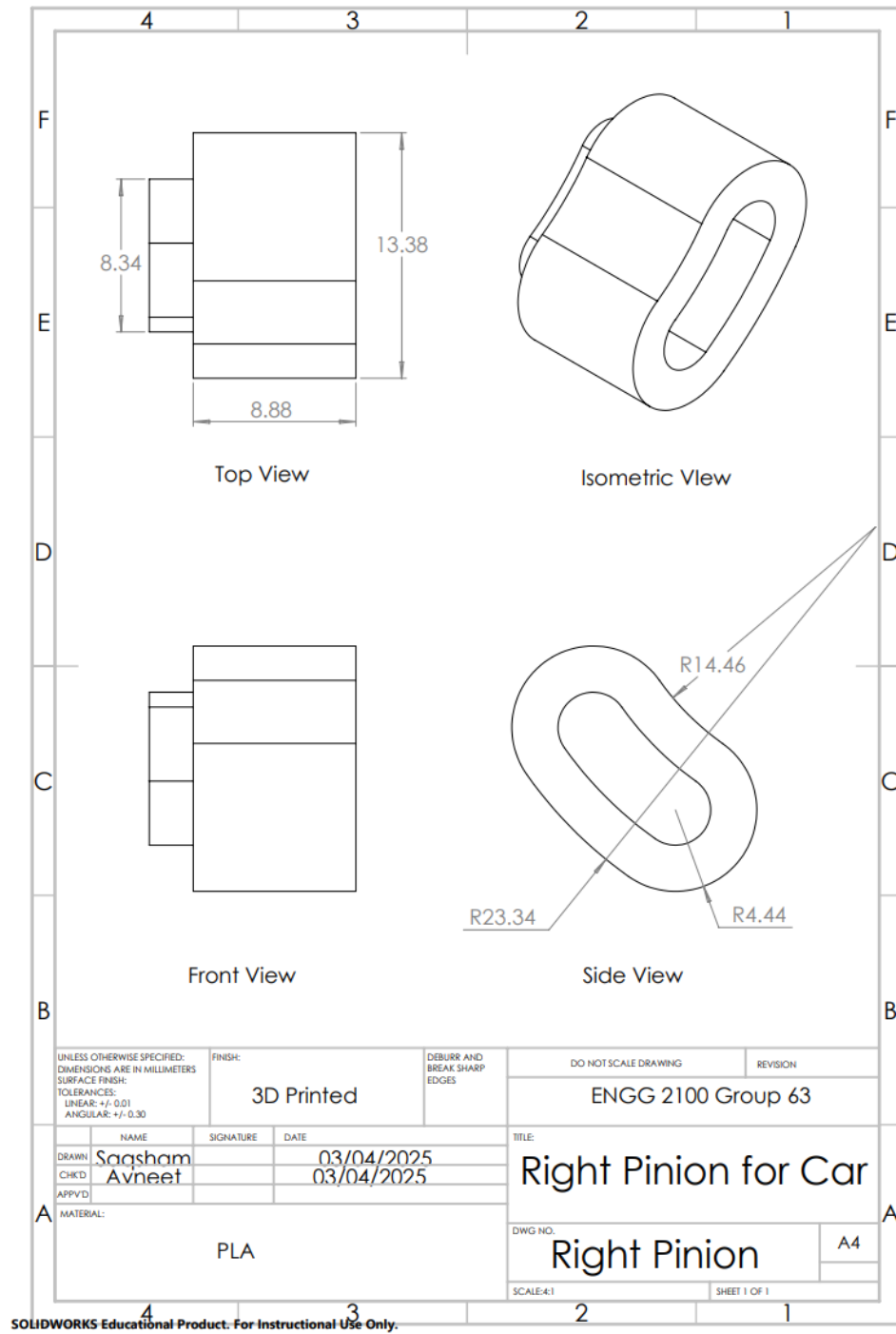


Figure 20: 2D Drawing – Right Pinion

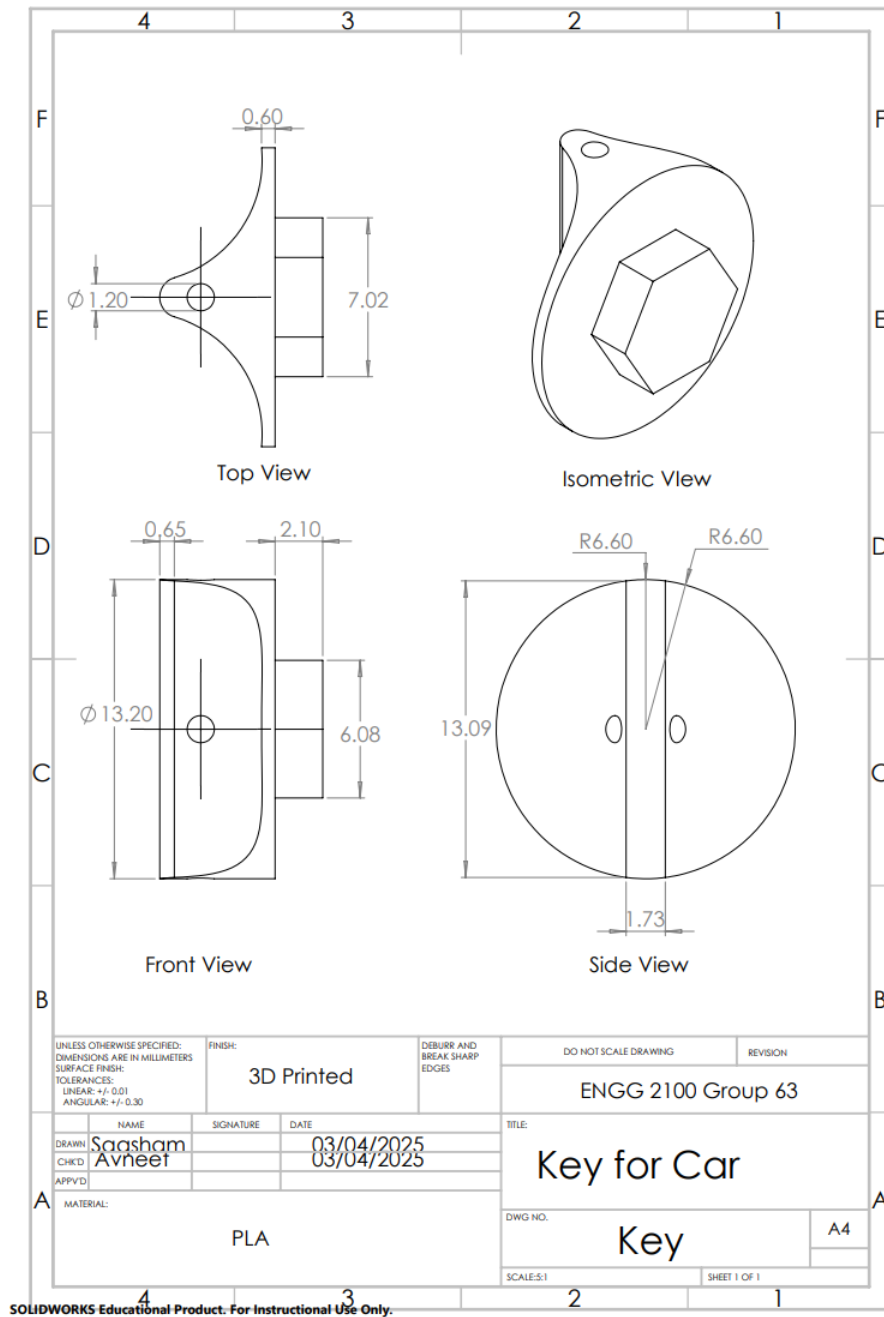


Figure 210: 2D Drawing – Key

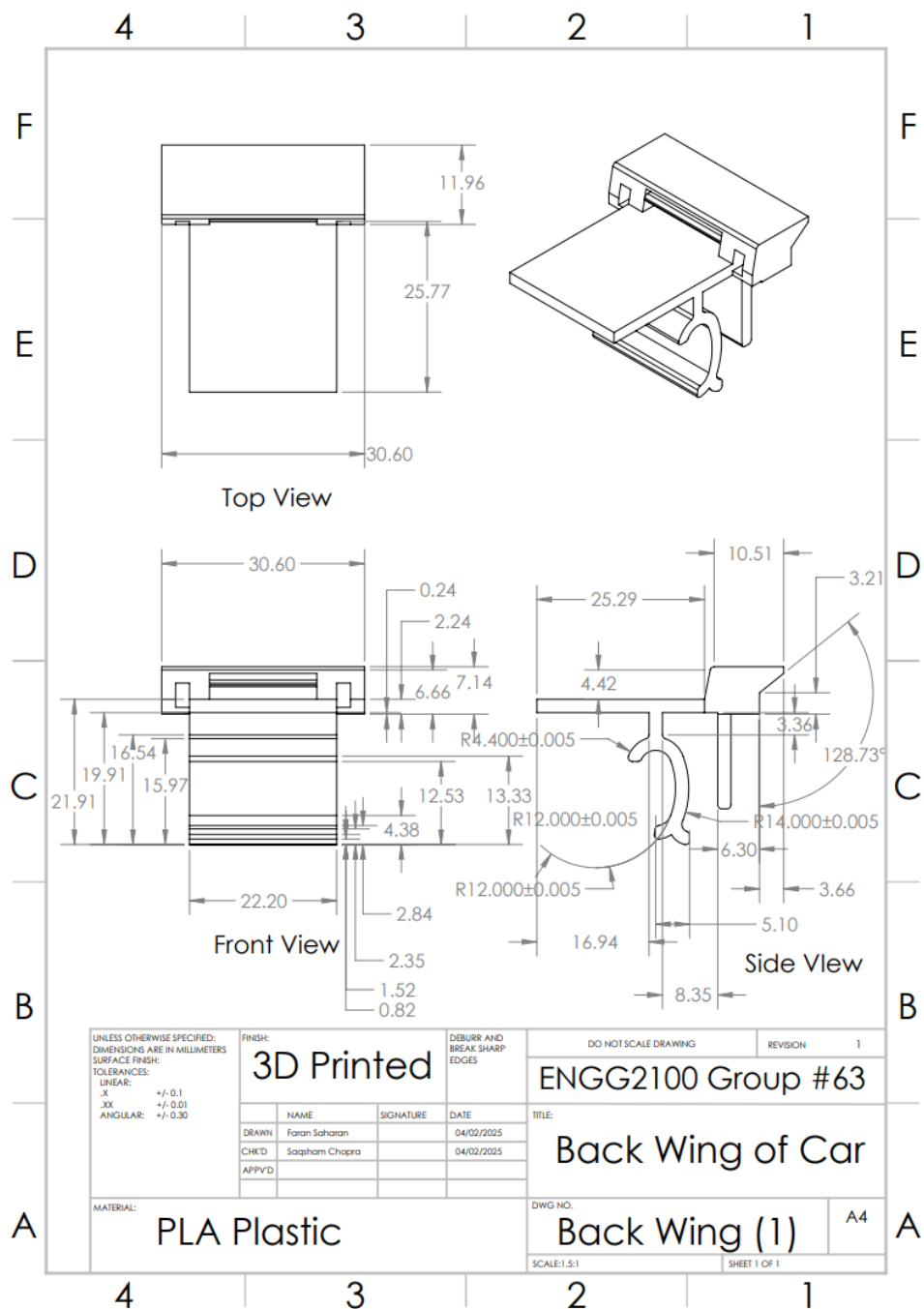


Figure 112: 2D Drawing – Back Wing

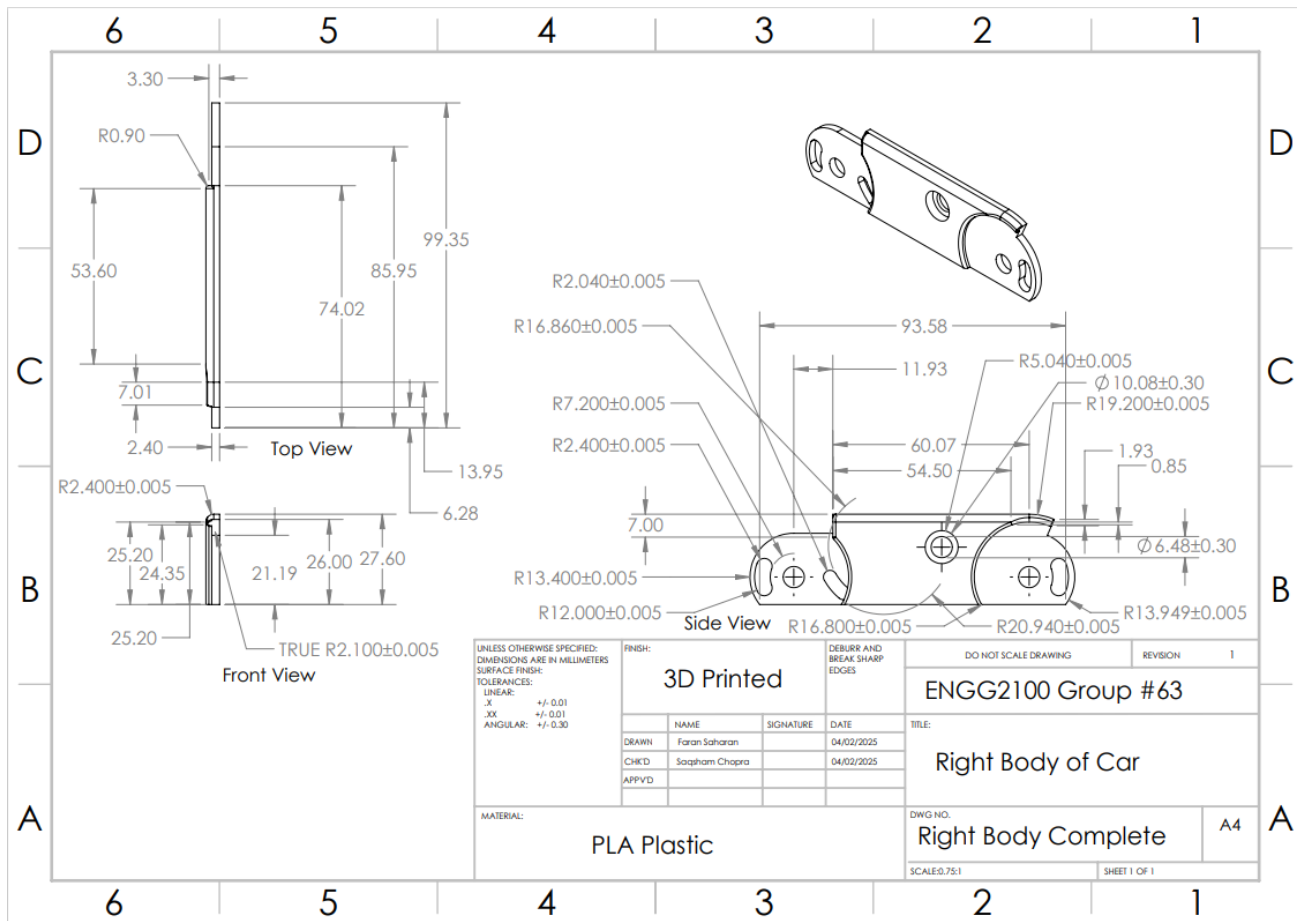


Figure 12: 2D Drawing – Right Body Complete

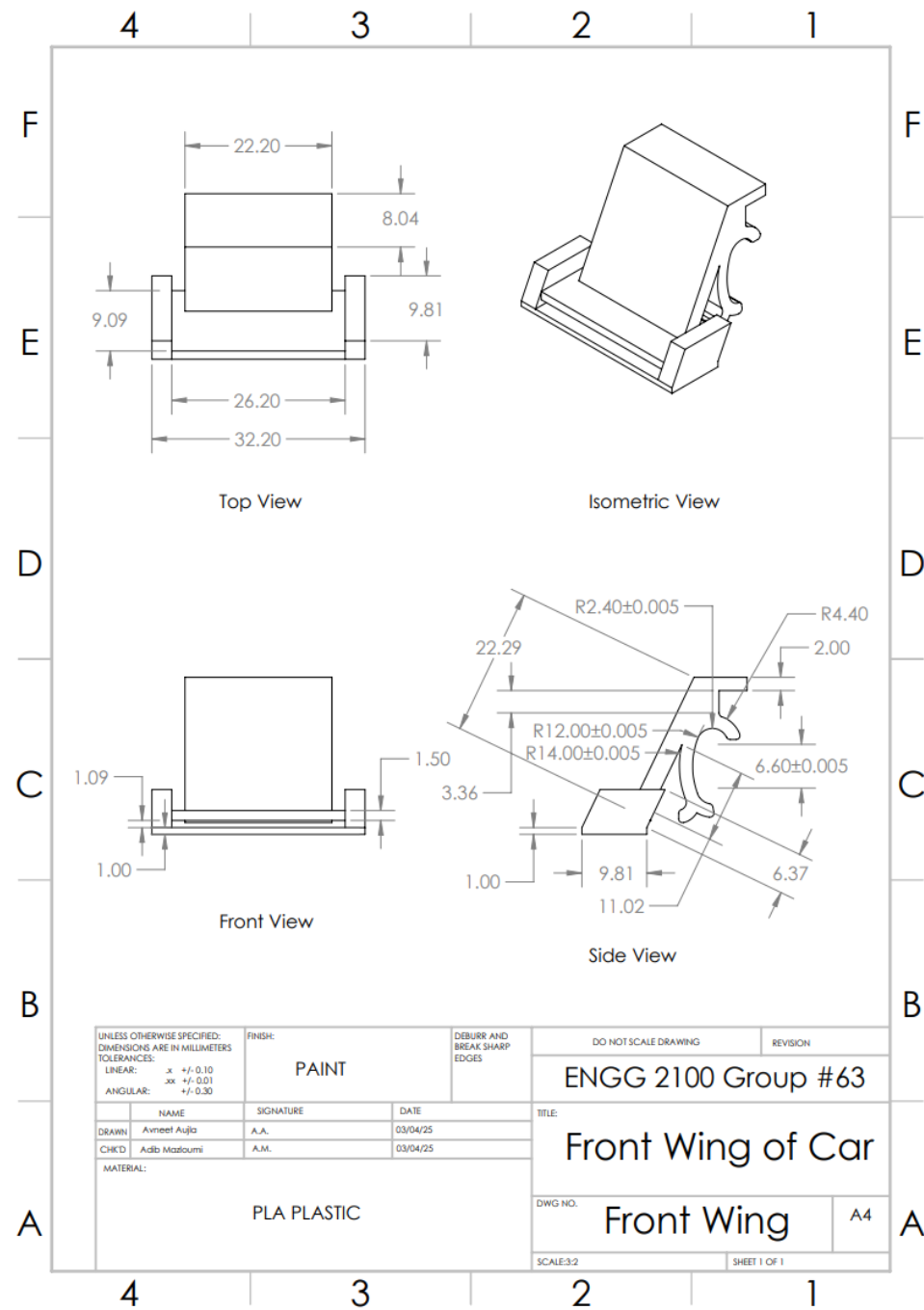


Figure 13: 2D Drawing – Front Wing

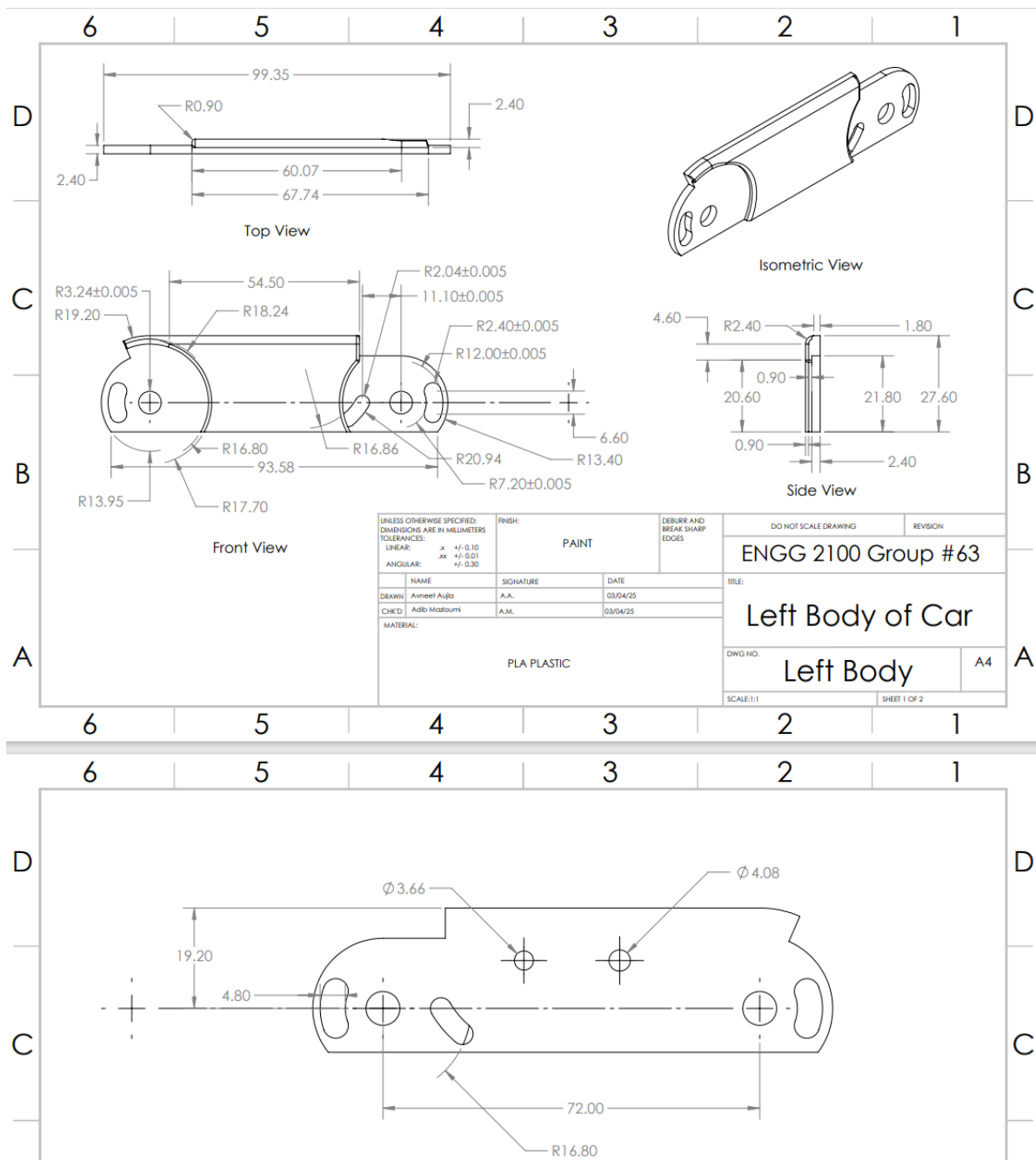


Figure 14: 2D Drawing – Left Body

Appendix C



Figure 15: Example Assembly instructions for Blueprint 1 Toy Car

Appendix D

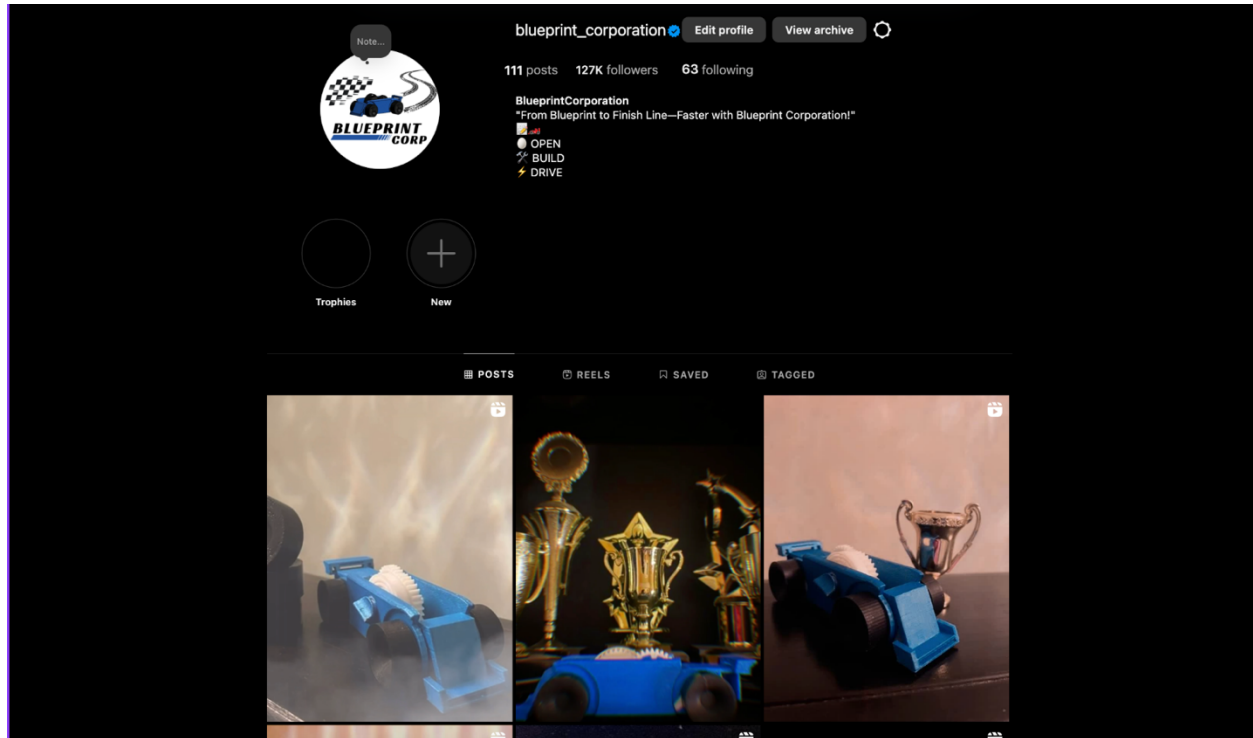


Figure 16: Company Social Media Page

Appendix E

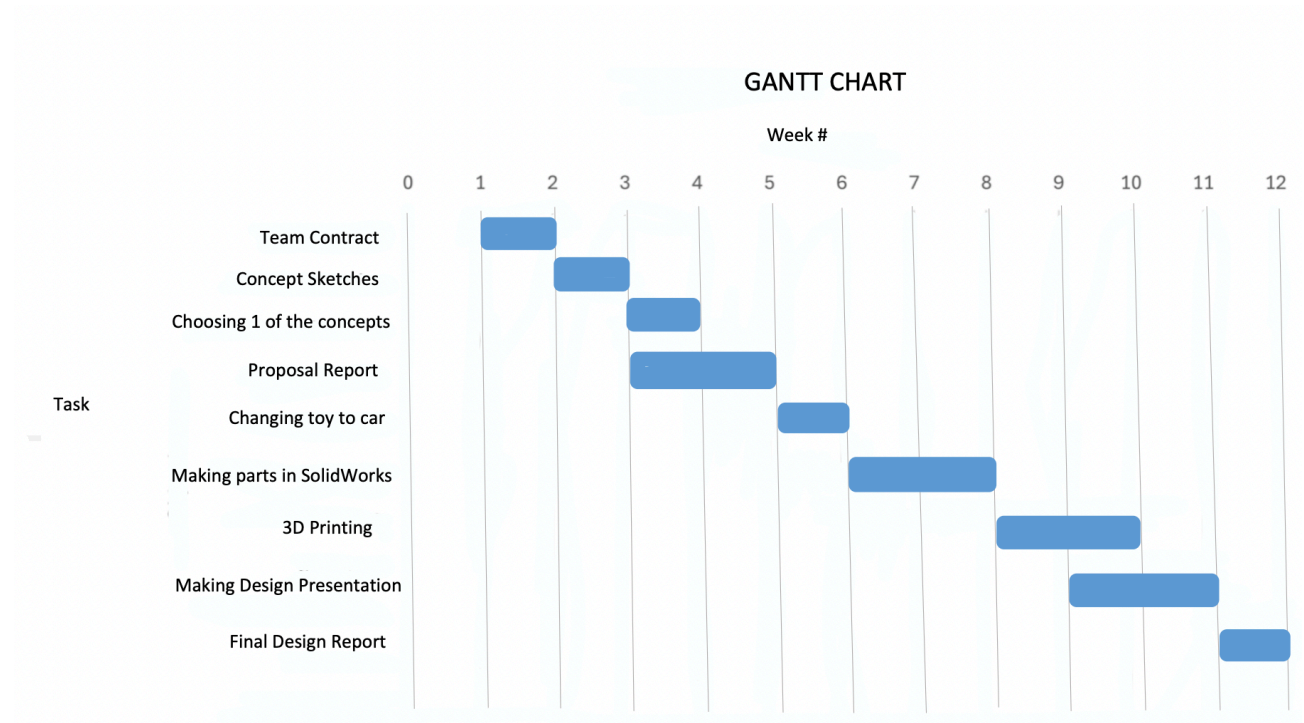


Figure 17: Gantt Chart